

Verification as a Costed Action: When Checking Beats Containing Under Attack–Fault Ambiguity

Chiheb Mejrissi*

*Department of Computer Science, Example University, 1 Example Street, Example City,
Country*

Abstract

An operator who sees only order flow cannot always tell whether a trading-gateway anomaly comes from a compromised account or a benign fault. Verification, such as asking the account holder to confirm through a separate channel, reads authorization and is not bound by what order flow can reveal. We model verification as a costed action with sensitivity, specificity, a fixed cost and a delay during which a true compromise keeps doing harm. Adapting clinical threshold analysis, we show that verification pays exactly on an interval of beliefs around the passive containment threshold: a closed-form, two-threshold leave / verify / contain rule. Because harm during the delay accrues only under compromise, the interval vanishes at a closed-form critical delay: a time budget beyond which no check should be run. The interval does not depend on the prior; the prior sets how much of a population’s posterior mass falls inside it, and the saving vanishes as compromise becomes rare. On the cells of a companion study in which passive classification does no better than the best constant policy, an exploratory test—specified after that study’s cost table had been examined, so not pre-registered—supports the prediction that targeted verification beats that policy over the region of (s, t, c_v, τ) that the two-threshold rule predicts from the cells’ corrected posteriors. Targeted verification is, however, not uniformly the cheapest policy: in several cells where passive classification succeeds it is more costly than the passive classifier, which points to miscalibrated corrected posteriors there.

Keywords: costly verification, value of information, threshold decision rule,

*Corresponding author.

Email address: `chiheb.mejrissi@example.org` (Chiheb Mejrissi)

1. Introduction

An operator of an electronic trading gateway sees an account submit an unusual pattern of orders. The pattern may come from an adversary who has taken over the account or the session, or from a benign malfunction of the client’s own systems. The operator can *contain*—suspend the session, pull the orders—or *leave* trading running. Containing a fault disrupts a legitimate client for nothing; leaving a compromise lets the adversary keep trading. When the two causes produce order flow that looks the same, no amount of order-flow analysis settles the matter.

The separate channel. A companion study, Paper 1 [1], makes this precise. In its linear-Gaussian order-flow model, when the confounding condition holds, order-flow observation cannot separate a compromise from a fault; the result of Paper 1 used here is its Proposition 1, by which an operator who sees only order flow can then do no better than act on the prior, and pays a cost of ambiguity, over an operator who knows the cause, equal to the expected regret of the better of the two terminal actions taken at the prior (restated in Appendix B, and for our loss table as Corollary 1(i)). Verification reads another channel, *authorization*, and, if its outcome is conditionally independent of order flow given the cause (Assumption 2), it is not subject to that limit; Section 3.4 develops the argument.

When a check pays. A check is not free. It costs analyst time [2, 3] and the friction of a re-authentication challenge for the client [4], and it takes time during which a real adversary keeps trading. We model verification as an action with sensitivity s , specificity t , a fixed cost c_v and a delay τ during which a true compromise causes damage at rate r_K , on top of a general loss table (C_K, C_F, H_K, H_F) for the two terminal actions. The passive operator contains when the belief in a compromise exceeds a threshold q^* (Lemma 1). The value of a check is a tent peaking at q^* (Lemma 3), and the check pays exactly where the tent exceeds its cost. The result is a leave / verify / contain rule with two thresholds $q_1 < q^* < q_2$ in closed form (Theorem 1). This rule is the threshold approach of Pauker and Kassirer [5] transposed to attack–fault triage (Section 2).

What the setting adds is the shape of the cost. Because harm during the delay accrues only if the cause is a compromise, the expected cost of checking, $c_v + q r_K \tau$, grows with the belief. As the delay grows, the interval (q_1, q_2) shrinks and vanishes at a critical delay τ^* (Theorem 2). Its use here is operational: τ^* is a time budget for a verification procedure. A check that takes longer than τ^* should never be run, at any belief.

What the prior changes, and what it does not. The verification interval is a set of beliefs; it does not depend on the prior probability of compromise. What depends on the prior is how much of the posterior mass of a population of cases falls inside it, and hence how much targeted verification saves on that population. We define these per-population quantities and show that the saving vanishes as the prior goes to zero, and so does the mass in the interval unless the check is both free and never flags a fault and the order flows of compromise and fault overlap (Lemma 4). A low prior therefore does not by itself make verification more valuable; whether a check helps in a given population is an empirical question about where that population’s posteriors fall relative to q_1 .

Contributions.

1. *Verification under a separate-channel assumption.* If the outcome of a check that reads authorization is conditionally independent of order flow given the cause (Assumption 2, a modelling assumption, not a result), the check composes with any passive posterior (Lemma 2) and lowers the confounded cost of ambiguity exactly when the prior lies in the verification interval (Corollary 1); the confounded case is the worst case for the passive and the verify-first operator alike, and verification caps the cost of ambiguity for every order-flow model (Corollary 2).
2. *The critical delay.* A closed-form τ^* at which the verification interval vanishes, with nested intervals below it and the manner of vanishing characterised (Theorem 2), under a cost of delay borne only by a true compromise; charging that cost in full makes τ^* a lower bound on the critical delay under any other accounting of the delay (Corollary 3).
3. *An exploratory empirical test* on the companion study’s market designs and priors (Section 5), with the data, the grid of (s, t, c_v, τ) , the plausible set, the five policies and the criteria all fixed before any result. It is exploratory because it was specified after the companion study’s cost table had been examined. In the cells where passive classification fails,

its outcome supports the prediction that targeted verification beats the best constant policy over the region the two-threshold rule predicts; across all cells, the policy comparison weakens the expectation that targeted verification is never beaten, since in several cells where passive classification succeeds the passive classifier is cheaper.

Section 2 positions the paper. Section 3 sets out the model and the separate-channel assumption, and Section 4 derives the rule, the critical delay and the per-population quantities; proofs are in Appendix A. Section 5 specifies the empirical design and reports its exploratory results. Section 6 discusses scope and limitations, and Section 7 concludes.

2. Related work

Threshold analysis and the value of information. The leave / verify / contain rule of Section 4 is the threshold approach of Pauker and Kassirer [5] transposed to attack–fault triage. There, a clinician’s probability of disease is split by a *testing threshold* and a *test-treatment threshold* into three regions: withhold treatment, test and follow the result, treat without testing. The thresholds follow from the test’s sensitivity and specificity, its cost or risk, and the benefit and harm of treatment. Our q_1 and q_2 play the same roles, with contain in place of treat and a general loss table (C_K, C_F, H_K, H_F) in place of the clinical utilities. The underlying calculation—a single experiment is worth running exactly when its expected value of sample information exceeds its cost, and for two actions and two states that value is piecewise linear and vanishes wherever no outcome can change the action—is classical Bayesian decision theory [6]. Fixing a test’s sensitivity and specificity when its output is continuous narrows the range of beliefs over which testing pays [7].

Costs of delay that depend on the state. The feature that distinguishes our setting is a cost of checking that falls on one cause: an adversary keeps trading while the check runs, while a fault is taken not to get worse (Assumption 4, relaxed in Corollary 4). Costs of delay that depend on the belief about the underlying state are a central object of time-critical decision analysis, where the expected cost of deferring an action is computed from a distribution over states and a time-dependent utility [8, 9]. In medical decision making, Schroeder et al. [10] use the threshold framework to trade the accuracy of a slower send-out test against the clinical benefit lost while waiting for its

result; this is the closest precedent we know for the delay term $qr_K\tau$ of our model. At the population level, the value of further research is likewise eroded by the decisions made while it is pending [11]. In operations, the value of diagnostic information is weighed against the delay that acquiring it imposes on the cases waiting, and decision makers set this trade-off poorly [12]. Given the tent, the critical delay τ^* of Theorem 2 follows in one step. Neither the rule, nor the tent shape of the value of information (Lemma 3), nor the critical delay is new as decision theory; what this paper adds is their use for attack–fault triage, where the cost of delay falls only on a compromise, and the separate-channel argument of Section 3.4.

Costly and active information acquisition. Choosing which observations to buy before acting is studied as Bayesian active learning and adaptive diagnosis, where tests are selected sequentially to identify a hypothesis at least cost [13], and as optimal stopping with costly sequential information, where the reversibility of the terminal decision shapes when to stop [14]. The value of information has also been characterised for stopping problems in which the decision maker pays for signals before an irreversible action [15]. Our problem is the one-shot special case with two causes, two terminal actions and a single available check. We keep it one-shot on purpose: it gives the thresholds and the critical delay in closed form, and the sequential version is a separate problem.

Intrusion response and alert verification. Intrusion response has long been cast as a cost-sensitive decision, in which the damage an attack would cause is weighed against the cost of responding [16], and later as a game or a partially observed control problem [17, 18, 19]. Alert verification checks whether an intrusion attempt actually succeeded before an alert is raised [20]; it is a technical filter, with no cost, delay or threshold attached. More recently, investigating an alert has been modelled as costly information acquisition that improves the defender’s belief [21, 22]; deferring uncertain alerts to an analyst is learned in the triage of security operations centres [23]; and a two-threshold rule escalates high scores, closes low ones and defers the rest [24]. There the thresholds come from error-rate budgets or are learned; the cost of the check does not grow with the belief, and there is no critical delay. Separating cyberattacks from physical faults has been studied through active detection [25] and, recently, by adding cyberspace logs to physical measurements, an analysis that in industry is manual and slow [26]. That

second line is the closest precedent we know for reading a second channel; what it does not give is a rule for when reading that channel is worth its cost and delay. Our contribution sits between these lines: a verify-then-respond decision layer with an explicit accuracy, cost and cause-borne delay, justified by the fact that the check reads a channel the observation-side analysis does not cover.

Prior shift and cost-sensitive classification. The empirical design of Section 5 uses classifier scores as passive posteriors. Scores trained at one class balance are moved to another prior by the standard correction [27], and a calibrated posterior is turned into a cost-minimising decision by thresholding it at the cost ratio [28], a rule also used with instance-dependent costs in fraud detection [29]. A cost-ratio threshold is optimal only for calibrated scores [30]; the passive threshold q^* of Lemma 1 is that threshold for our loss table.

3. Model

An operator watches an anomaly on a trading gateway and must decide what caused it. Two causes are possible: a *compromise* K , in which someone the account holder did not authorize submits orders through the holder’s valid credential or session (a stolen credential, a hijacked session), and a *fault* F , in which the orders are the holder’s own and the same symptoms come from a benign malfunction. The operator can *contain* (suspend the account or session, pull the orders) or *leave* (let trading continue). Before committing to either, the operator may also *verify*: query a source of evidence that the order flow does not carry. This section sets out the decision problem; Section 4 solves it.

3.1. Loss table and passive risk

The operator’s belief that the cause is a compromise is $q \in [0, 1]$. Without verification, the operator picks the terminal action with the lower expected loss.

Definition 1 (Loss table, passive risk). There are two causes, compromise K (the orders are submitted through a valid credential or session by someone the holder of that credential or session did not authorize: a stolen credential, a hijacked session) and fault F (the orders are the holder’s own, distorted by a benign malfunction), and two terminal actions, *contain* and

leave. Containing costs C_K under K and C_F under F ; leaving costs H_K under K and H_F under F . For a belief $q = P(K) \in [0, 1]$ the *passive risk* is

$$R(q) = \min\{q C_K + (1 - q) C_F, q H_K + (1 - q) H_F\}.$$

Write $a = H_K - C_K$ and $b = C_F - H_F$.

The four entries of the loss table are kept general. Containing a compromise (C_K) costs the disruption of a legitimate client plus the operational cost of the response; containing a fault (C_F) costs the same disruption with nothing gained; leaving a compromise (H_K) costs the harm the adversary goes on to do; leaving a fault (H_F) costs whatever the malfunction costs if it is not interrupted, counted from the terminal decision on; what a malfunction costs during the check itself is the rate r_F of Corollary 4, zero under Assumption 4. The quantity a is what containment saves when the cause is a compromise, and b is what it wastes when the cause is a fault. The cost matrix of Paper 1 [1] is the special case $H_F = 0$, with Paper 1's containment cost, disruption and harm in the roles of C_K , C_F and H_K .

Nothing in the model depends on the prior probability of compromise except through the belief q . The prior of a single case enters in Section 4.6 (Definition 4 and Corollary 1); the distribution of passive posteriors over a population of cases enters in Section 4.8 (Definition 5).

3.2. Verification

Verification is a costed action with an uncertain outcome. On a trading gateway it is, for example, a call from the desk to the client to confirm the activity, or a step-up authentication that asks the submitter for a factor bound to the account holder. A check that the credential or session is valid is not a verification in this sense: a stolen credential or a hijacked session is valid.

Definition 2 (Verification action). Verification returns an outcome $Z \in \{+, -\}$ with sensitivity $s = P(Z = + \mid K)$ and specificity $t = P(Z = - \mid F)$. It costs c_v and takes a delay τ ; during the delay a true compromise causes harm at rate r_K . From belief q ,

$$P(+) = qs + (1 - q)(1 - t), \quad P(-) = q(1 - s) + (1 - q)t,$$

$$q^+ = \frac{qs}{P(+)}, \quad q^- = \frac{q(1 - s)}{P(-)}$$

(a posterior whose outcome has probability zero is never used; see Definition 3).

The sensitivity s is the probability that verification flags a compromise that is present; the specificity t is the probability that it clears a fault. The cost c_v is paid whatever the cause: analyst time, the call, the friction imposed on the client. The delay τ is the time the check takes, and the damage rate r_K is the harm per unit of that time done by an adversary who is really present (a rate, distinct from the cost H_K of leaving a compromise). We assume that a fault accrues no additional cost while the operator is checking, whereas a compromise does (Assumption 4); Corollary 4 drops this assumption.

Definition 3 (Verify-first risk, value of information, verification region). The *verify-first risk* is

$$R_v(q) = c_v + q r_K \tau + P(+) R(q^+) + P(-) R(q^-),$$

where $P(z) R(q^z)$ is read as $\min\{P(z, K) C_K + P(z, F) C_F, P(z, K) H_K + P(z, F) H_F\}$ with $P(z, K) = P(z \mid K) q$ and $P(z, F) = P(z \mid F) (1 - q)$ (equal to the product when $P(z) > 0$, and zero when $P(z) = 0$). The *value of information* is $V(q) = R(q) - [P(+) R(q^+) + P(-) R(q^-)]$, the *verification cost* is $\chi(q) = c_v + q r_K \tau$, and the *verification region* is

$$S = \{q \in [0, 1] : R_v(q) < R(q)\} = \{q \in [0, 1] : V(q) > \chi(q)\}.$$

Ties are resolved in favour of the passive action: the operator verifies only when verifying is strictly cheaper.

The verify-first risk adds two terms to the expected passive risk at the updated belief: the cost of the check and the harm expected during the delay. The value of information V is the familiar reduction in terminal risk that the outcome buys [6]; the verification cost χ collects what the check itself costs. Because harm during the delay accrues only under K , χ grows with the belief: checking is more expensive precisely when a compromise is more likely. The verification region S is the set of beliefs at which the check pays for itself.

3.3. Assumptions

Assumption 1 (Cost orderings). $H_K > C_K$ and $C_F > H_F$; that is, $a > 0$ and $b > 0$: leaving a compromise is strictly worse than containing it, and containing a fault is strictly worse than leaving it.

This excludes the cases in which one terminal action dominates the other whatever the cause, and the case $a, b < 0$, which is the same problem with

the names of the two actions exchanged (Remark 1(b)). On a gateway it says that an unchecked adversary does more damage than a response costs, and that suspending a healthy client costs more than letting a benign fault run. Remark 1 in Section 4 records what happens when it fails.

Assumption 2 (Separate channel). Verification reads authorization, not order flow: it asks whether the holder of the credential or session authorized the orders, through a channel the person submitting them does not control (for example an out-of-band confirmation with the account holder, or a step-up authentication with a factor bound to the holder). A check of whether the credential or session is valid is not such a check, since a stolen credential or a hijacked session is valid under K . Its outcome Z is conditionally independent of the order-flow observation Y given the cause: $P(Z, Y \mid c) = P(Z \mid c)P(Y \mid c)$ for $c \in \{K, F\}$, and (s, t) do not depend on Y .

Assumption 2 is the conceptual core of the paper and is discussed in Section 3.4.

Assumption 3 (Informative one-shot verification). $s, t \in [0, 1]$ with $s + t > 1$; $c_v \geq 0$, $\tau \geq 0$, $r_K \geq 0$. Verification is available once; after its outcome the operator takes a terminal action (contain or leave) and cannot verify again.

The condition $s + t > 1$ says that a positive outcome is more likely under a compromise than under a fault; a check that fails it carries no information (Remark 1). The one-shot restriction matches a triage step in which one check is run and the operator then acts. Repeated or sequential checking is a stopping problem and is outside the scope of this paper.

Assumption 4 (Delay accounting). The harm $r_K \tau$ accrued by a true compromise during the verification delay is *additional* to the terminal cost (C_K or H_K) incurred afterwards, whichever terminal action is taken; the terminal costs C_K, C_F, H_K, H_F and the verification characteristics s, t do not depend on τ ; a fault accrues no additional cost during the delay (its damage rate is $r_F = 0$); and no further order-flow evidence is used between the start of verification and the terminal action.

Assumption 4 fixes how the delay is charged. The harm accrued during the check is paid in addition to the terminal cost, whatever the operator then does; this is the conservative accounting for verification, since it charges the delay in full even when the terminal cost H_K might already include part of it. Any double counting is therefore against verification only: under any accounting in which H_K already includes part or all of that harm, the critical

delay τ^* of Theorem 2 is a lower bound (Corollary 3). The check does not change the loss table or its own accuracy, a fault costs nothing extra while it is checked (relaxed in Corollary 4), and the operator does not update on further order flow while waiting for the outcome.

3.4. *When verification is not bound by the order-flow limit*

In its linear-Gaussian order-flow model, when the confounding condition holds, Paper 1 [1] shows that order-flow observation cannot separate a compromise from a fault, and that a passive operator who sees only order flow then pays a cost of ambiguity that no classifier can remove. That limit is a statement about the law of the order-flow observation Y under the two causes; it constrains every decision rule that is a function of Y alone. It does not constrain evidence that is not a function of Y .

Verification is such evidence. Such a check reads the authorization channel: whether the entity submitting the orders is the one entitled to submit them. Assumption 2 states that, given the cause, the outcome of that check is independent of what the order flow shows, and that its accuracy (s, t) does not depend on the order flow. Operationally, whether the account holder confirms the orders, or the submitter passes a challenge bound to the holder, does not depend on how the orders look. The sensitivity s is the probability that the holder disavows the orders, or that the submitter fails the challenge, under a compromise; the specificity t is the probability that the holder confirms them under a fault. The check does not detect a holder who trades maliciously on their own account, nor an adversary who also controls the holder’s second factor; the model leaves both out of K . Under this assumption verification composes with whatever the passive analysis concludes: Lemma 2 in Section 4 shows that the joint posterior is the verification update applied to the order-flow posterior. That the characteristics (s, t) are the same whatever the order-flow law is not a finding of the lemma but Assumption 2 itself; what the lemma adds is that, given the assumption, the two updates compose, including when the order-flow laws under K and F coincide. That is the case in which Paper 1’s passive operator can do no better than act on the prior; Corollary 1 in Section 4 states when verification brings the cost of ambiguity strictly below Paper 1’s value there.

Assumption 2 is a modelling choice, not a consequence of the setting; Section 6.2 states what is lost when it fails.

4. The two-threshold rule and the critical delay

This section derives when verification pays. The passive rule has a single threshold (Lemma 1). Verification composes with any order-flow posterior (Lemma 2), and its value is a tent peaking at that threshold (Lemma 3). Comparing the tent with the cost of the check gives a leave / verify / contain rule with two thresholds (Theorem 1), and a critical delay beyond which the check never pays (Theorem 2). Corollary 1 applies these results to the confounded case, Corollary 2 shows that this case is the worst case, Corollaries 3 and 4 show how the critical delay depends on the accounting of the delay, and Remark 1 records the degenerate parameters. Lemma 4 and Proposition 1 then turn from a single belief to a population of cases with a given prior, which is where the prior enters. Proofs are in Appendix A.

4.1. The passive threshold

Lemma 1 (Passive threshold). Under Assumption 1, let $q^* = b/(a + b) \in (0, 1)$. Leaving is strictly optimal for $q < q^*$, containing is strictly optimal for $q > q^*$, and

$$R(q) = q H_K + (1 - q) H_F - (a + b) (q - q^*)^+,$$

so R is continuous, concave and piecewise linear on $[0, 1]$ with a single kink at q^* .

The operator who cannot verify contains exactly when the belief in a compromise exceeds q^* : the share of the total stakes $a + b$ that a wrong containment wastes. A response that is costly to a healthy client (large b) pushes q^* up; an adversary who does much damage if left alone (large a) pulls it down. At q^* the two terminal actions are equally good, and the passive risk has its only kink.

4.2. Verification composes with the passive posterior

Lemma 2 (Verification composes with any passive posterior). Under Assumption 2, let $q(y) = P(K \mid Y = y)$ be the passive (order-flow) posterior. Then $P(K \mid Y = y, Z = z)$ equals the verification update of Definition 2 applied to $q = q(y)$: it is $q(y)^+$ if $z = +$ and $q(y)^-$ if $z = -$. In particular the characteristics (s, t) , and hence V , S and all results below, are the same whatever the order-flow law, including when the order-flow laws under K and F coincide.

Lemma 2 separates the two stages of the decision. The operator first reduces the order-flow evidence to a belief $q(y)$, with whatever classifier or model is available, and then applies the verification analysis below to that belief, with the same (s, t) , which Assumption 2 posits rather than Lemma 2 derives. The verification rule need not be re-derived for each classifier, and it remains meaningful when the order flow carries no information about the cause (Section 3.4). All results below are stated for a belief q and hold at $q = q(y)$.

4.3. The value of information is a tent

Lemma 3 (The value of information is a tent). Under Assumptions 1 and 3,

$$V(q) = \max\{0, \min\{G^+(q), G^-(q)\}\},$$

where

$$G^+(q) = q s a - (1 - q)(1 - t) b, \quad G^-(q) = (1 - q) t b - q(1 - s) a.$$

Consequently, with

$$q_L = \frac{(1 - t) b}{s a + (1 - t) b}, \quad q_U = \frac{t b}{t b + (1 - s) a},$$

(i) $0 \leq q_L < q^* < q_U \leq 1$; (ii) $V(q) > 0$ if and only if $q \in (q_L, q_U)$, equivalently if and only if $q^- < q^* < q^+$ (the two posteriors fall strictly on opposite sides of q^*); (iii) V is affine and strictly increasing on $[q_L, q^*]$, affine and strictly decreasing on $[q^*, q_U]$, and zero outside (q_L, q_U) ; (iv) V attains its maximum uniquely at q^* , with

$$V(q^*) = (s + t - 1) \frac{a b}{a + b} = (s + t - 1) a q^* > 0.$$

Lemma 3 is the standard value-of-information calculation for two actions and two states [6]. A check is worth something only when its outcome can change the terminal action: a positive outcome must move the belief above q^* and a negative outcome below it. Outside the interval (q_L, q_U) neither outcome can overturn the passive decision, and the check is worthless there. That interval widens as the check becomes more accurate: a perfect check ($s = t = 1$) has $q_L = 0$ and $q_U = 1$, and has value at every belief strictly between 0 and 1. The value rises linearly from q_L to its peak at q^* and falls

linearly to q_U , with slopes that differ in general, so the tent is not symmetric about q^* (Figure 1). The height of the peak is the product of the Youden index $s + t - 1$ and the quantity $a q^* = b(1 - q^*)$, which is the expected regret at the threshold, relative to an operator who knows the cause, of either terminal action.

4.4. The two-threshold rule

Theorem 1 (Two-threshold leave / verify / contain rule). Under Assumptions 1, 3 and 4, the verification region S is an open interval (possibly empty). It is non-empty if and only if

$$V(q^*) > c_v + q^* r_K \tau,$$

and then $S = (q_1, q_2)$ with $q_L \leq q_1 < q^* < q_2 \leq q_U$, where

$$q_1 = \frac{c_v + (1 - t)b}{s a + (1 - t)b - r_K \tau}, \quad q_2 = \frac{t b - c_v}{t b + (1 - s)a + r_K \tau}$$

(both denominators are positive when $S \neq \emptyset$). The optimal one-shot policy is: *leave* if $q \leq q_1$, *verify* if $q_1 < q < q_2$ and then contain on $Z = +$ and leave on $Z = -$, *contain* if $q \geq q_2$ (with $q_1 = q_2 = q^*$ and the passive rule of Lemma 1 when $S = \emptyset$). When $c_v = 0$ and $\tau r_K = 0$, $S = (q_L, q_U)$.

Theorem 1 is the threshold rule of Pauker and Kassirer [5], with q_1 and q_2 in the roles of the testing and test-treatment thresholds (Section 2); what the transposition adds is the cost of the check that grows with the belief, through the delay term $q r_K \tau$.

For an operator the rule reads as follows. Below q_1 the check costs more than it is expected to save, and the operator leaves. Above q_2 a compromise is likely enough that the operator contains without checking. In between, the operator verifies and follows the outcome (Figure 1). The verification interval always contains the passive threshold q^* when it is non-empty, and it lies inside the interval (q_L, q_U) on which the check has any value at all. Its two ends move inward as the check becomes more expensive or, for $\tau < \tau^*$, slower: c_v raises the lower end and lowers the upper one, and the delay term $r_K \tau$ enters both denominators.

4.5. The critical delay

Theorem 2 (Critical delay). Under Assumptions 1, 3 and 4, write $S(\tau)$ for the verification region at delay τ , all other parameters fixed. (i)

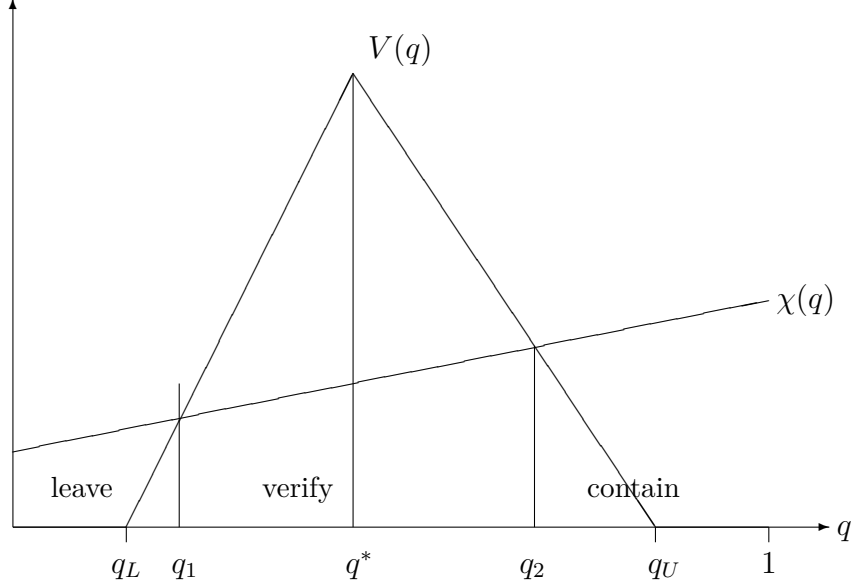


Figure 1: Schematic of Lemma 3 and Theorem 1 (not to scale; no parameter values). The value of information V is a tent on (q_L, q_U) with its peak at the passive threshold q^* ; the verification cost χ increases with the belief because harm during the delay accrues only under compromise. Verification pays on (q_1, q_2) , where the tent lies above the cost line. Raising the delay τ tilts the cost line upward and shrinks the interval (Theorem 2).

Nesting. $\tau < \tau'$ implies $S(\tau') \subseteq S(\tau)$. If $r_K > 0$ and $V(q^*) > c_v$, then $S(\tau) \neq \emptyset$ exactly for $\tau \in [0, \tau^*)$ (part (ii)), and on $[0, \tau^*)$ the end point $q_2(\tau)$ of Theorem 1 is strictly decreasing in τ , and $q_1(\tau)$ is strictly increasing in τ unless $c_v = 0$ and $t = 1$, in which case $q_1(\tau) = 0$ on $[0, \tau^*)$. For $\tau \geq \tau^*$, $S(\tau) = \emptyset$ and the convention $q_1 = q_2 = q^*$ of Theorem 1 applies. (ii) *Critical delay.* If $r_K > 0$ and $V(q^*) > c_v$, let

$$\tau^* = \frac{V(q^*) - c_v}{q^* r_K} = \frac{(s + t - 1)a - c_v/q^*}{r_K} > 0.$$

Then $S(\tau) \neq \emptyset$ if and only if $\tau < \tau^*$. (iii) *Degenerate cases.* If $V(q^*) \leq c_v$, then $S(\tau) = \emptyset$ for every $\tau \geq 0$. If $r_K = 0$, $S(\tau)$ does not depend on τ . (iv) *How the interval vanishes.* Suppose $r_K > 0$ and $V(q^*) > c_v$ (the hypotheses of (ii)). If $c_v > 0$ or $t < 1$, then $q_1 \rightarrow q^*$ and $q_2 \rightarrow q^*$ as $\tau \uparrow \tau^*$: the interval shrinks continuously to the point q^* . If $c_v = 0$ and $t = 1$, then $S(\tau) = (0, q_2(\tau))$ for $\tau < \tau^*$ with $q_2(\tau) \downarrow q^*$, and $S(\tau^*) = \emptyset$: the interval vanishes discontinuously.

Because the value of information is a tent peaking at q^* and the cost of the check is affine and increasing in the belief, the check pays at some belief if and only if it pays at q^* , and solving that inequality for the delay gives τ^* (precedents in Section 2).

For an operator, τ^* is a time budget for the check. This reading assumes that a fault costs nothing during the check (Assumption 4); if it does, the budget is the shorter τ_F^* of Corollary 4. A verification procedure that takes longer than τ^* should never be run, at any belief: the operator should act on the passive rule instead. Below τ^* the verification interval shrinks as the check slows down, and the operator verifies over a narrower band of beliefs around q^* . The budget grows with the accuracy of the check (through $s + t - 1$) and shrinks with the damage rate r_K and with the fixed cost c_v . If the fixed cost alone is at least the peak value of information, $c_v \geq V(q^*)$, no delay is short enough: the check is never worth running (Theorem 2(iii)). In the limiting case of a free check that never flags a fault ($c_v = 0$ and $t = 1$), the interval reaches down to zero belief and disappears all at once at τ^* ; otherwise it closes continuously onto q^* .

4.6. Comparison with the passive cost of ambiguity

The results above are stated for a belief. To compare verification with the passive analysis of Paper 1 [1], the decision problem is stated over policies that act on the order-flow observation.

Definition 4 (Decision problem with verification; verify-first cost of ambiguity). The cause M is K with prior probability $p = P(K) \in (0, 1)$ and F otherwise. Given M , the order-flow observation Y has law P_M^Y , and the verification outcome Z has the law of Definition 2, jointly as in Assumption 2. Write $L(\text{contain}, K) = C_K$, $L(\text{contain}, F) = C_F$, $L(\text{leave}, K) = H_K$, $L(\text{leave}, F) = H_F$ (Definition 1). A *verify-first policy* is a quadruple of measurable maps $\delta, \pi_0, \pi_+, \pi_-$ from observations y to $[0, 1]$: with probability $\delta(y)$ the operator verifies and then contains with probability $\pi_Z(y)$; otherwise it contains with probability $\pi_0(y)$. Its expected cost is

$$\begin{aligned} J(\delta, \pi) = \mathbb{E} \bigg[& (1 - \delta(Y)) (\pi_0(Y) L(\text{contain}, M) \\ & + (1 - \pi_0(Y)) L(\text{leave}, M)) \\ & + \delta(Y) (c_v + \mathbb{E}[M = K] r_K \tau + \pi_Z(Y) L(\text{contain}, M) \\ & + (1 - \pi_Z(Y)) L(\text{leave}, M)) \bigg]. \end{aligned}$$

A *passive policy* is a verify-first policy with $\delta \equiv 0$. Let R_Y^* and $R_{Y,v}^*$ be the infima of J over passive and over verify-first policies, $R_{\text{oracle}} = p \min\{C_K, H_K\} + (1-p) \min\{C_F, H_F\}$ the expected cost of an operator who observes M , and

$$\text{CoA}^* = R_Y^* - R_{\text{oracle}}, \quad \text{CoA}_v^* = R_{Y,v}^* - R_{\text{oracle}}$$

the passive and the verify-first cost of ambiguity.

Corollary 1 (Verification lowers the confounded cost of ambiguity). Under Assumptions 1–4, in the setting of Definition 4, suppose the order-flow laws coincide, $P_K^Y = P_F^Y$ (the confounded case). Then: (i) $\text{CoA}^* = \min\{p a, (1-p) b\} > 0$; (ii) the verify-first cost of ambiguity is

$$\text{CoA}_v^* = \min\{\text{CoA}^*, E(p)\},$$

$$E(p) = c_v + p r_K \tau + p(1-s)a + (1-p)(1-t)b,$$

equivalently $\text{CoA}_v^* = \text{CoA}^* - (V(p) - \chi(p))^+$, and it is attained by the policy of Theorem 1 applied at the belief $q = p$, whatever Y ; (iii) $\text{CoA}_v^* < \text{CoA}^*$ if and only if $p \in S$, that is, if and only if $V(q^*) > c_v + q^* r_K \tau$ and $q_1 < p < q_2$ (Theorem 1); if $r_K > 0$ and $V(q^*) > c_v$, this holds if and only if $\tau < \tau^*$ and $q_1(\tau) < p < q_2(\tau)$ (Theorem 2), and if $V(q^*) \leq c_v$ it holds for no p and no τ ; (iv) over $p \in (0, 1)$, the reduction $\text{CoA}^* - \text{CoA}_v^*$ is largest at $p = q^*$, where it equals $(V(q^*) - c_v - q^* r_K \tau)^+$; (v) $\text{CoA}_v^* = 0$ if and only if $s = t = 1$, $c_v = 0$ and $r_K \tau = 0$. Under the correspondence of Paper 1, Definition D8 ($H_F = 0$), CoA^* in (i) is the cost of ambiguity of Paper 1, Proposition 1, so (iii) states when verification brings the cost of ambiguity strictly below that value.

In the confounded case the order flow carries no information about the cause, and the passive cost of ambiguity in part (i), stated for the general loss table, reduces to the value of Paper 1's Proposition 1 when $H_F = 0$. Corollary 1 states when verification brings it strictly lower: exactly when the prior lies in the verification region of Theorem 1, so, for a harmful compromise and a check whose fixed cost is below the peak value of information, only when the check is faster than τ^* . The reduction is largest when the prior sits at the passive threshold, and the cost of ambiguity is removed entirely only by a perfect and free check that accrues no harm during its delay. The optimal verify-first policy ignores the order flow, which here carries no information, and applies the rule of Theorem 1 to the prior.

Corollary 2 (Uniform ceiling: the confounded case is the worst case). Under Assumptions 1–4, in the setting of Definition 4, let $E(p)$ be as

in Corollary 1. For every pair of order-flow laws P_K^Y, P_F^Y and every $p \in (0, 1)$: (i) the *verify-everything* policy ($\delta \equiv 1, \pi_+ \equiv 1, \pi_- \equiv 0$) has expected cost $R_{\text{oracle}} + E(p)$, whatever the order-flow laws; (ii) $\text{CoA}^* \leq \min\{p a, (1-p) b\}$ and

$$\text{CoA}_v^* \leq \min\{\text{CoA}^*, E(p)\} \leq \min\{p a, (1-p) b, E(p)\};$$

(iii) every inequality in (ii) is an equality in the confounded case $P_K^Y = P_F^Y$ (Corollary 1), so that, over all pairs of order-flow laws, the largest passive cost of ambiguity is $\min\{p a, (1-p) b\}$ and the largest verify-first cost of ambiguity is $\min\{p a, (1-p) b, E(p)\}$, both attained in the confounded case; (iv) the largest verify-first cost of ambiguity is strictly below the largest passive one if and only if $p \in S$, and the gap is $(V(p) - \chi(p))^+$. Under the correspondence of Paper 1, Definition D8 ($H_F = 0$), and the hypotheses of Paper 1, Proposition 1, $\min\{p a, (1-p) b\}$ is the cost of ambiguity of Paper 1, Proposition 1, and the passive bound in (ii) is the upper bound of Paper 1, Proposition 2 (total-variation extension of Proposition 1): the value of Paper 1's Proposition 1 is the worst case of the passive cost of ambiguity over all order-flow models, and verification caps the cost of ambiguity at $E(p)$ for every order-flow model.

Corollary 2 says that the confounded case is the worst case for the passive and for the verify-first operator alike: whatever the order-flow model, verification caps the cost of ambiguity at $E(p)$, the cost of ambiguity of the policy that verifies every case.

Corollary 3 (Delay accounting: τ^* is a lower bound). For $\theta \in [0, 1]$, *Assumption 4 $_\theta$* is Assumption 4 except that the harm $r_K \tau$ accrued by a true compromise during the delay is added to the terminal cost in full if the terminal action is *contain* and in the share θ if it is *leave*. Assumption 4₁ is Assumption 4; Assumption 4₀ is the accounting in which H_K already includes the harm a compromise left in place causes during the delay. Under Assumption 4 $_\theta$ the verify-first risk is

$$R_v^\theta(q) = c_v + \sum_{z \in \{+, -\}} \min\left\{P(z, K) (C_K + r_K \tau) + P(z, F) C_F, \right. \\ \left. P(z, K) (H_K + \theta r_K \tau) + P(z, F) H_F\right\},$$

with $P(z, K), P(z, F)$ as in Definition 3, and the verification region is $S_\theta(\tau) = \{q \in [0, 1] : R_v^\theta(q) < R(q)\}$. Let $\sigma_\theta = s + \theta(1-s) \in [s, 1]$. Under Assumptions 1, 3 and 4 $_\theta$: (i) $S_\theta(\tau) = \{q \in [0, 1] : V(q) > c_v + \sigma_\theta q r_K \tau\}$, the verification

region of Definition 3 with r_K replaced by $\sigma_\theta r_K$; for $q \in S_\theta(\tau)$ the optimal action after verification is to contain on $Z = +$ and leave on $Z = -$. Theorems 1 and 2 therefore hold under Assumption 4 $_\theta$ with r_K replaced by $\sigma_\theta r_K$. (ii) For $0 \leq \theta \leq \theta' \leq 1$ and every $\tau \geq 0$, $S(\tau) = S_1(\tau) \subseteq S_{\theta'}(\tau) \subseteq S_\theta(\tau) \subseteq S_0(\tau)$. (iii) If $r_K > 0$ and $V(q^*) > c_v$, then $S_\theta(\tau) \neq \emptyset$ if and only if $\tau < \tau_\theta^*$, where

$$\tau_\theta^* = \frac{\tau^*}{\sigma_\theta}, \quad \tau^* \leq \tau_\theta^* \leq \frac{\tau^*}{s},$$

with τ^* of Theorem 2. So τ^* is a lower bound on the critical delay under every accounting 4 $_\theta$, attained if and only if $\theta = 1$ or $s = 1$.

Corollary 3 answers the objection that the harm done during the delay is counted twice when H_K already includes it: charging that harm in full, as Assumption 4 does, only shortens the critical delay, so τ^* is a lower bound on the critical delay under every such accounting.

Corollary 4 (A fault that accrues cost during the delay). For $r_F \geq 0$, Assumption 4 $_F$ is Assumption 4 except that a fault also accrues cost at rate r_F during the verification delay, additional to the terminal cost whichever terminal action is taken (Assumption 4 is the case $r_F = 0$). Under Assumption 4 $_F$ the verify-first risk is $R_v(q) + (1 - q)r_F\tau$, so the verification region is that of Definition 3 with χ replaced by

$$\chi_F(q) = c_v + qr_K\tau + (1 - q)r_F\tau = (c_v + r_F\tau) + q(r_K - r_F)\tau.$$

Under Assumptions 1, 3 and 4 $_F$: (i) S is an open interval, non-empty if and only if $V(q^*) > c_v + \bar{r}\tau$ with $\bar{r} = q^*r_K + (1 - q^*)r_F$, and then $S = (q_1, q_2)$ with $q_L \leq q_1 < q^* < q_2 \leq q_U$,

$$q_1 = \frac{c_v + r_F\tau + (1 - t)b}{sa + (1 - t)b - (r_K - r_F)\tau}, \quad q_2 = \frac{tb - c_v - r_F\tau}{tb + (1 - s)a + (r_K - r_F)\tau},$$

both denominators positive; the optimal policy is that of Theorem 1 with these end points. (ii) $\tau < \tau'$ implies $S(\tau') \subseteq S(\tau)$. If $V(q^*) \leq c_v$, then $S(\tau) = \emptyset$ for every $\tau \geq 0$. If $\bar{r} > 0$ and $V(q^*) > c_v$, then $S(\tau) \neq \emptyset$ if and only if $\tau < \tau_F^*$, where

$$\tau_F^* = \frac{V(q^*) - c_v}{q^*r_K + (1 - q^*)r_F}.$$

(iii) If moreover $r_K > 0$, then $\tau_F^* = \tau^* q^* r_K / \bar{r} \leq \tau^*$, with equality if and only if $r_F = 0$: the critical delay τ^* of Theorem 2 is an upper bound on the critical delay when faults accrue cost during the delay.

If a fault also costs during the check, the verification region is still an interval and vanishes at a critical delay $\tau_F^* \leq \tau^*$; so τ^* is the time budget when a fault costs nothing during the check and an upper bound on it otherwise, while Corollary 3 shows it is a lower bound when H_K already includes the delay harm.

4.7. Degenerate parameters

Remark 1 (Degenerate parameters). (a) If $s+t = 1$, then $q^+ = q^- = q$ whenever defined, $V \equiv 0$, and $S = \emptyset$ for all $c_v \geq 0$, $\tau \geq 0$. (b) Assumption 1 fails exactly when $a \leq 0$ or $b \leq 0$, that is, in one of two cases. (b1) *Weak signs mixed or zero:* $a \leq 0 \leq b$ or $b \leq 0 \leq a$ (so $q^* = b/(a+b) \notin (0,1)$ or is undefined). Then one terminal action is weakly optimal at every belief, R is affine, $V \equiv 0$, and $S = \emptyset$. (b2) *Both negative:* $a < 0$ and $b < 0$. Then $q^* = b/(a+b) \in (0,1)$, containing is strictly optimal for $q < q^*$ and leaving for $q > q^*$. Exchanging the names of the two terminal actions gives a problem with the same R , V , χ and S that satisfies Assumption 1 with (a,b) replaced by $(-a,-b)$ and the same q^* ; every result below then holds for that problem, i.e. for the original one with (a,b) replaced by $(-a,-b)$ and *contain* and *leave* exchanged (in Theorem 1: contain if $q \leq q_1$, verify if $q_1 < q < q_2$ and then leave on $Z = +$ and contain on $Z = -$, leave if $q \geq q_2$). (c) The requirement $c_v \geq 0$ in Assumption 3 is needed for Theorem 1: if $c_v < 0$, S can contain beliefs near 0 where $V = 0$ and can then be disconnected.

Parts (a) and (b) say that the results above need a check that is informative and a decision that is not settled in advance. Part (c) explains why Assumption 3 requires a non-negative cost: a check that pays the operator to run it would be worth running at beliefs where its outcome cannot change the action, and the verification region could then fail to be an interval.

4.8. Populations of cases: what the prior changes

The verification region $S = (q_1, q_2)$ is a set of beliefs. Its end points depend on the loss table and on (s, t, c_v, τ, r_K) , and not on the prior probability of compromise. An operator facing a population of cases—for instance all the anomalies raised in one simulated market at one base rate of compromise—sees instead a distribution of passive posteriors, and the prior moves that distribution. The following quantities measure how much of it falls in the verification region and what verifying there saves.

Definition 5 (Verification mass, saving and recovered share of a cell). An *order-flow model* c is a pair of order-flow laws P_K^c, P_F^c on a

measurable space, with densities f_K^c, f_F^c with respect to a σ -finite measure ν dominating both (e.g. $\nu = P_K^c + P_F^c$). A *cell* (c, p) is an order-flow model c together with a prior $p \in (0, 1)$; in Paper 1's grid, c is a market design in one regime and p is a prior of the grid. In the setting of Definition 4 with $P_K^Y = P_K^c, P_F^Y = P_F^c$ and prior $p \in (0, 1)$, write $P_p = p P_K^c + (1 - p) P_F^c$ for the law of Y and

$$q_p(y) = \frac{p f_K^c(y)}{p f_K^c(y) + (1 - p) f_F^c(y)}$$

for the passive posterior (defined where the denominator is positive, a set of full P_p -measure). The *verification mass*, the *saving* and the *recovered share* of the cell (c, p) are

$$m_c(p) = P_p(q_p(Y) \in S), \quad D_c(p) = \mathbb{E}_p[(V - \chi)^+(q_p(Y))],$$

$$\rho_c(p) = \frac{D_c(p)}{\text{CoA}_c^*(p)},$$

with S, V, χ as in Definition 3 and $\text{CoA}_c^*(p)$ the passive cost of ambiguity of Definition 4 for this cell; $\rho_c(p)$ is defined when $\text{CoA}_c^*(p) > 0$. Where $f_K^c, f_F^c > 0$, $\text{logit } q_p(y) = \text{logit } p + \Lambda_c(y)$ with $\Lambda_c = \log(f_K^c/f_F^c)$.

Lemma 4 (Saving identity and low-prior limit). Under Assumptions 1–4, in the setting of Definition 5: (i) *Saving.* $D_c(p) = \text{CoA}_c^*(p) - \text{CoA}_{v,c}^*(p) = R_Y^* - R_{Y,v}^*$, where $\text{CoA}_{v,c}^*(p)$ is the verify-first cost of ambiguity of Definition 4 for this cell; it is attained by the policy that verifies iff $q_p(Y) \in S$, then contains on $Z = +$ and leaves on $Z = -$, and otherwise takes the passive action of Lemma 1 at $q_p(Y)$. Moreover $0 \leq D_c(p) \leq \min\{\text{CoA}_c^*(p), V(q^*) m_c(p)\}$, so $0 \leq \rho_c(p) \leq 1$ where defined. (ii) *The saving vanishes at low priors.* For every order-flow model c , $D_c(p) \rightarrow 0$ as $p \rightarrow 0$. (iii) *The mass vanishes at low priors when $q_1 > 0$.* If $S = \emptyset$, then $m_c \equiv 0$. If $S = (q_1, q_2) \neq \emptyset$ and $q_1 > 0$ (equivalently, $c_v > 0$ or $t < 1$), then $m_c(p) \rightarrow 0$ as $p \rightarrow 0$. (iv) *The exception.* If $S \neq \emptyset, c_v = 0$ and $t = 1$ (so $q_1 = 0$), then $m_c(p) \rightarrow P_F^c(f_K^c(Y) > 0)$ as $p \rightarrow 0$. This limit is positive if and only if P_K^c and P_F^c are not mutually singular, and zero when they are (compromise and fault perfectly separable by order flow); the saving still vanishes, by (ii). No condition on the order-flow model is needed beyond Definition 5: the densities are finite, and $f_F^c > 0$ holds P_F^c -almost surely by construction.

Part (i) identifies the saving of targeted verification with the reduction in the cost of ambiguity, and bounds it by the peak value of information

times the mass in the interval. Parts (ii) and (iii) state what happens when compromise becomes rare. As the prior falls, the passive posteriors move towards zero and, in the limit, below the lower threshold q_1 whenever the check has a fixed cost or can flag a fault, so the mass in the interval and the saving both vanish. A low prior therefore does not make verification more valuable on its own: what matters is whether the upper tail of the posteriors still reaches above q_1 . Part (iv) is the only exception for the mass: for a free check that never flags a fault $q_1 = 0$ by the formula of Theorem 1, so the interval reaches down to zero belief, and the mass tends to $P_F^c(f_K^c(Y) > 0)$, which is positive exactly when the order-flow laws of compromise and fault are not mutually singular, although the saving still vanishes by part (ii).

Lemma 4 is the reason the empirical question of Section 5 is posed per population. Whether targeted verification saves anything in a given population depends on where that population's posteriors fall relative to q_1 , which the theory does not decide. For a Gaussian order-flow model the quantities of Definition 5 have closed forms.

Proposition 1 (Closed forms for a Gaussian order-flow model).

Under Assumptions 1–4, let the order-flow model c of Definition 5 be Gaussian: $Y \mid M \sim \mathcal{N}(\mu_M, \Sigma_Y)$, $M \in \{K, F\}$, with common covariance $\Sigma_Y \succ 0$ and whitened separation $d = \sqrt{(\mu_K - \mu_F)^\top \Sigma_Y^{-1} (\mu_K - \mu_F)} > 0$. Suppose $S = (q_1, q_2) \neq \emptyset$. Write $\ell = \text{logit } p$, $\ell_1 = \text{logit } q_1$, $\ell^* = \text{logit } q^* = \log(b/a)$, $\ell_2 = \text{logit } q_2$ (with $\text{logit } 0 = -\infty$, $\text{logit } 1 = +\infty$, $\Phi(\pm\infty) = 1, 0$ accordingly), and for $-\infty \leq u < v \leq +\infty$

$$\begin{aligned}\Pi_K(u, v) &= \Phi\left(\frac{v - \ell}{d} - \frac{d}{2}\right) - \Phi\left(\frac{u - \ell}{d} - \frac{d}{2}\right), \\ \Pi_F(u, v) &= \Phi\left(\frac{v - \ell}{d} + \frac{d}{2}\right) - \Phi\left(\frac{u - \ell}{d} + \frac{d}{2}\right).\end{aligned}$$

Then

$$\begin{aligned}m_c(p) &= p \Pi_K(\ell_1, \ell_2) + (1 - p) \Pi_F(\ell_1, \ell_2), \\ D_c(p) &= p(\alpha_1 - \beta_1) \Pi_K(\ell_1, \ell^*) - (1 - p) \beta_1 \Pi_F(\ell_1, \ell^*) \\ &\quad + p(\beta_2 - \alpha_2) \Pi_K(\ell^*, \ell_2) + (1 - p) \beta_2 \Pi_F(\ell^*, \ell_2),\end{aligned}$$

with $\alpha_1 = s a + (1 - t) b - r_K \tau$, $\beta_1 = c_v + (1 - t) b$, $\alpha_2 = t b + (1 - s) a + r_K \tau$, $\beta_2 = t b - c_v$ (so that $q_1 = \beta_1/\alpha_1$ and $q_2 = \beta_2/\alpha_2$, Theorem 1). As $d \rightarrow 0$, $m_c(p) \rightarrow \mathbb{1}[p \in S]$ for $p \notin \{q_1, q_2\}$ and $D_c(p) \rightarrow (V - \chi)^+(p)$ for every p , the confounded values of Corollary 1.

Proposition 1 shows how the mass and the saving depend on the separation d and on the prior through the logit shift ℓ , and it recovers the confounded values of Corollary 1 as $d \rightarrow 0$.

The order-flow models of Paper 1 [1], to which Remark 2 and Section 5 refer, are simulated with ABIDES, an agent-based discrete-event simulator of a limit order book market [31]. In Paper 1 a *market design* is one way of injecting a compromise and a fault into the simulated market, and each design is run in two *regimes*: in the *summary-matched* regime the two injections are parameterised so that summary statistics of their order flow match, and in the *contrast* regime they are not. A design in one regime is an order-flow model c in the sense of Definition 5.

Remark 2 (The Gaussian model is a calibration model). The hypotheses of Proposition 1 are a Gaussian order-flow model of the form of Paper 1’s Definition D11 with any separation $d > 0$; they involve no value of d and no property of Paper 1’s cells. Paper 1’s ABIDES order-flow models were never shown to be Gaussian, so the Gaussian model is an *equivalent-Gaussian calibration model*, not Paper 1’s cells, and Proposition 1 describes those cells only insofar as they are Gaussian with that separation. A separation d backed out of a Paper 1 accuracy through the Gaussian benchmark $\text{Acc}^* = \Phi(d/2)$ is a calibration heuristic only; wherever the closed forms of Proposition 1 are evaluated at such a d , d is also swept directly over a grid. Every statement about Paper 1’s cells, including where targeted verification beats the best constant policy in them, rests on quantities computed on each cell’s corrected classifier posteriors (Definition 5 applies to any order-flow model with finite densities), with no Gaussian assumption, and not on Proposition 1.

The benchmark $\text{Acc}^* = \Phi(d/2)$ of Remark 2 is the accuracy of the Bayes classifier of the Gaussian model at equal priors, $p = 1/2$. At a prior p with $\ell = \text{logit } p$, the Bayes accuracy of the same model is $p \Phi(d/2 + \ell/d) + (1 - p) \Phi(d/2 - \ell/d)$, which exceeds $\Phi(d/2)$ whenever $p \neq 1/2$; a separation backed out through $\Phi(d/2)$ from an accuracy measured at another prior therefore overstates d . This is a further reason why such a d is a calibration heuristic only and why d is also swept directly over a grid.

5. Exploratory empirical study on Paper 1’s cells

This section specifies an empirical study on the order-flow models and priors of Paper 1 [1]. The study is exploratory: its primary test was specified after Paper 1’s cost table for the same cells had been examined, so

whatever it finds is exploratory evidence, not the outcome of a pre-registered test. Everything below—data, cells, grids, plausible set, policies, metrics and outcome criteria—was fixed before any quantity of the primary test was computed. The theorems of Section 4 are proved and are not tested here; where the study computes a quantity that a theorem gives in closed form, a mismatch is an error of computation, not evidence.

5.1. Data and passive posteriors

The order-flow data are Paper 1’s simulated order-flow features for each of its market designs, in both the summary-matched and the contrast regime (Section 4.8). On each design and regime, Paper 1’s classifier of compromise against fault is refitted exactly as Paper 1 fits it, on Paper 1’s primary training split, and applied to the held-out test instances, which are balanced between the two causes. Fits that Paper 1’s procedure leaves non-converged are kept as they are, because the cells studied are defined by Paper 1’s own pipeline; convergence is recorded per design and regime.

The classifier’s scores are estimated at the class balance of the training set, with share π_0 of compromises. They are moved to a prior p by the standard prior-shift correction [27]: a score r becomes the corrected posterior

$$q_p(r) = \frac{(p/\pi_0) r}{(p/\pi_0) r + ((1-p)/(1-\pi_0))(1-r)}.$$

The corrected posteriors are the passive beliefs $q(y)$ to which Lemma 2 applies the verification analysis.

5.2. Cells and the failure cells

A cell is a design, a regime and a prior (Definition 5): the design in one regime is the order-flow model c , and p ranges over the priors of Paper 1’s grid. Every quantity of Definition 5 is an expectation under $P_p = p P_K^c + (1-p) P_F^c$. On the balanced test set this law is obtained by weighting each compromise instance by p/n_K and each fault instance by $(1-p)/n_F$, with n_K and n_F the numbers of test instances of each cause; quantiles of the posterior are lower quantiles of this weighted empirical distribution. Without the weights, the test set would describe the posteriors under the training balance, which is the prior the correction removes.

The *failure cells* are the cells in which Paper 1 reports that its classifier does not beat the best constant policy, always contain or always leave. They

are read from Paper 1's reported cost table, with Paper 1's best constant policy for each cell, and are not re-derived here.

The loss table is Paper 1's cost matrix in the form of Section 3.1: $H_F = 0$ and Paper 1's containment cost, disruption and harm in the roles of C_K , C_F and H_K . This fixes a , b and the passive threshold $q^* = b/(a+b)$ of Lemma 1, which are the same in every cell.

5.3. Verification grid in model units

The model involves the delay and the damage rate only through their product $r_K\tau$ (Definition 2, Assumption 4), so the grid has four axes, written in units of the model. Let $V_{\max} = a q^* = ab/(a+b)$, the peak value of a perfect, free and instantaneous check (Lemma 3), which no check can exceed at any belief. The axes are the sensitivity s , the specificity t ,

$$x = \frac{c_v}{V_{\max}} \quad \text{and} \quad y = \frac{r_K\tau}{a},$$

the cost of a check as a share of the most any check can save, and the harm a true compromise accrues during the delay as a share of the harm cost difference a . In these units the condition of Theorem 1 for a non-empty verification interval reads $x + y < s + t - 1$, and Theorem 2 gives the critical delay as $r_K\tau^*/a = s + t - 1 - x$, so critical delays are reported in units of a/r_K . Whether the interval exists at a grid point therefore does not depend on the monetary scale of a cell. The same monetary values of c_v and $r_K\tau$ are used in every cell.

Sensitivity and specificity range over the full product of a common set of levels, every pair of which satisfies Assumption 3. The cost axis x and the delay axis y each range over a set of levels starting at a near-free check and at no delay respectively, and extending far enough that, for every pair (s, t) , the grid has points on both sides of the boundary $x + y = s + t - 1$ at which the interval vanishes (Theorem 2(ii)). The levels of every axis are listed with the results (Section 5.8).

Plausible set. A check is trivially useful at low beliefs when its lower threshold is near zero. By the formula for q_1 in Theorem 1, $q_1 = 0$ when the fixed cost c_v is zero and the specificity t is one, and a near-free check that almost never flags a fault pushes q_1 close to zero; the limiting case is Lemma 4(iv), in which the mass in the interval does not vanish at low priors. A grid point is therefore *implausible* if and only if its specificity is the highest level of the grid

and its cost is one of the two lowest levels of x , whatever its sensitivity and its delay; every other grid point is *plausible*. The sensitivity is not restricted, because it enters q_1 only through the factor sa in the denominator, which the grid bounds away from zero. The delay is not restricted, because the delay cost $qr_K\tau$ is scaled by the belief and cannot make a check near-free at low posteriors. Costly or slow checks are not implausible: they only shrink or empty the interval, which the criteria below count against the prediction.

5.4. Policies compared

Five policies are compared on every cell and grid point: always contain; always leave; the passive classifier, which contains when the corrected posterior exceeds q^* (Lemma 1); verify everything and follow the outcome; and targeted verification by the rule of Theorem 1 applied to the corrected posterior. Verification outcomes are not simulated. For a test instance with true cause M and posterior q , the cost of each policy is its exact expectation over the outcome Z , with $P(Z = + \mid K) = s$ and $P(Z = + \mid F) = 1 - t$ (Definition 2), plus the fixed cost and the harm during the delay where the policy verifies (Assumption 4). The only randomness left in the study is the resampling of test instances for the confidence intervals.

5.5. Metrics

Per cell and grid point the study computes:

- q_1 , q_2 and τ^* in closed form (Theorems 1 and 2), cross-checked against a numerical search of the rule;
- the location of the corrected posteriors: their quantiles under each cause, the shares below q_1 , inside (q_1, q_2) and above q_2 , and the distance of the median corrected logit posterior from $\text{logit } q^*$;
- the verification mass $m_c(p)$ of Definition 5, estimated as the P_p -weighted share of test instances whose corrected posterior lies in (q_1, q_2) ;
- the saving $D_c(p)$ of Definition 5, estimated in two ways: the *plug-in saving* $\widetilde{D}_c(p)$, the P_p -weighted mean of $(V - \chi)^+$ at the corrected posteriors, which is Definition 5 evaluated as if the corrected posteriors were the true ones; and the *realised saving* $\widehat{D}_c(p)$, the P_p -weighted cost of the passive rule minus that of targeted verification, computed from the true causes of the test instances. By Lemma 4(i) the two agree in expectation when the corrected posteriors are calibrated;

- the recovered share $\rho_c(p)$, estimated as $\widehat{D}_c(p)$ divided by the passive cost of ambiguity, itself estimated as the P_p -weighted cost of the passive rule minus the cost of an operator who knows the cause; it is not reported where that cost of ambiguity is indistinguishable from zero;
- for every policy, its expected cost and, beside it, its false-positive and false-negative rates.

False-positive and false-negative rates are reported beside every cost because, as in Paper 1, the prior-shift correction trades one for the other. Confidence intervals are bootstrap percentile intervals over test instances, resampled within each cause so that the class sizes, and hence the weights of Section 5.2, are exact in every resample. The confidence level and the number of resamples are listed with the results.

5.6. Primary exploratory test: targeted verification in the failure cells

The primary test asks, in each failure cell, over which region of (s, t, c_v, τ) targeted verification beats the cell's best constant policy, and whether that region is the one predicted from Theorem 1 and the cell's corrected posteriors.

Contrast and predicted saving. Write $\Delta(c, g)$ for the expected cost of targeted verification minus that of the best constant policy, in cell (c, p) at grid point g , with bootstrap confidence interval $[L, U]$. When the best constant policy is always leave, Definition 1 and Lemma 1 give that leaving costs $(a+b)(q-q^*)^+$ more than the passive rule at belief q , and Lemma 4(i) gives that targeted verification costs $(V-\chi)^+(q)$ less than it. The model's prediction for the tested contrast is therefore $-\pi(c, g)$, with predicted saving

$$\pi(c, g) = \mathbb{E}_p \left[(a+b)(q-q^*)^+ + (V-\chi)^+(q) \right],$$

the expectation taken over the corrected posteriors under P_p . When the best constant policy is always contain, $(q-q^*)^+$ is replaced by $(q^*-q)^+$. The two terms are reported separately; a positive first term in a cell where the passive rule does not beat the best constant policy indicates miscalibrated corrected posteriors and is reported as such.

Predicted region. By Theorem 1 and Lemma 4(i), a saving over always leaving can come only from posterior mass above q_1 . Let $Q_\alpha = \inf\{u : P_p(q \leq u) \geq \alpha\}$ be the α -quantile of the corrected posterior q under P_p , and fix two quantile levels $\alpha_{\text{in}} < \alpha_{\text{out}}$ in the upper tail. A grid point is

- *predicted in* if $q_1 < Q_{\alpha_{\text{in}}}$ and $P_p(q_1 < q < q_2) > 0$;
- *predicted out* if $q_1 \geq Q_{\alpha_{\text{out}}}$, or if all of the mass above q_1 lies at or above q_2 ;
- in the *band* otherwise.

The prediction speaks of the upper tail of the posteriors without naming a level. Outside the band, the position of q_1 relative to Q_α is the same at every level $\alpha \in [\alpha_{\text{in}}, \alpha_{\text{out}}]$, so the classification does not depend on the choice of level; band points are exactly those whose prediction would change with it, so they are counted and reported, with the quantile at an intermediate level, and enter no share. The two levels are set from the size of the test set, not from any result. At α_{in} the tail above q_1 holds enough weighted test instances for the bootstrap interval to resolve a saving of the predicted order, so a point is declared in only where the test can see the predicted saving; at α_{out} it holds too few instances for any interval to resolve one, and the model puts almost no mass where verification acts, so a point is declared out only there. The predicted region uses the point estimates on the full test set.

Agreement. At a predicted-in point the empirical result *agrees* if $U < 0$, *disagrees* if $U \geq 0$ and $L > -\pi$, and is *unresolved* if the interval contains both 0 and $-\pi$, which is a failure of power rather than a disagreement. At a predicted-out point it agrees if $U \geq 0$ and disagrees if $U < 0$, a saving the model does not predict. A cell's agreement share is the number of agreeing points divided by the number of agreeing and disagreeing points, over plausible grid points only; implausible points are classified and reported, and enter no share. A cell *agrees* if its share is at least $\bar{\sigma} = 1 - 2(1 - \gamma)$, with γ the confidence level of the intervals. If the prediction holds and the posteriors are calibrated, a point disagrees only through the error of its interval, at a rate of at most $1 - \gamma$; $\bar{\sigma}$ allows twice that rate, for the correlation of neighbouring grid points and residual miscalibration of the corrected posteriors. A cell is *unresolved* if more than half of its plausible predicted-in points are unresolved. Shares are decided per failure cell, because the predicted region is computed from each cell's own posteriors; a pooled share is reported and decides nothing. The values of α_{in} , α_{out} and γ are listed with the results.

Power. The size of the test set is fixed by Paper 1's design, so no sample size was chosen for a target power, and no power analysis in that sense was run.

Power is assessed instead point by point, against the effect the model itself predicts: at a predicted-in point the interval either separates the predicted saving π from zero or contains both 0 and $-\pi$, and in the second case the point is unresolved, not a disagreement. The outcome below uses this in both directions: a cell with a majority of unresolved predicted-in points makes the outcome *inconclusive at this power* rather than a support, and a refutation requires that the test had power where the model predicts a saving.

Outcome. The outcome is one of four, applied in this order.

1. *Refutes* if no plausible point of any failure cell has $U < 0$, and the test has power: at least one failure cell has plausible predicted-in points, and in every such cell at least half of them disagree. A cell with no predicted-in points predicts an empty region and cannot refute.
2. *Weakens* if some failure cell has an agreement share below $\bar{\sigma}$, or if $U < 0$ only at implausible points.
3. *Inconclusive at this power* if no cell weakens or refutes but some cell is unresolved.
4. *Supports* if every failure cell agrees, no cell is unresolved, and $U < 0$ at one plausible point of one failure cell at least.

A cell with no decided point is listed as such; it counts as unresolved if it has predicted-in points and otherwise enters no criterion. The outcome is decided with Paper 1’s cost matrix as above; a version with the measured cost of leaving a fault (Section 5.7) is reported beside it and decides nothing.

5.7. Secondary and descriptive analyses

Policy comparison. If the corrected posteriors are calibrated, targeted verification is, in expectation, at least as cheap as every other policy on every cell (Lemma 4(i) and the definitions of the constant policies). The comparison is read as weakening this if some other policy beats targeted verification beyond sampling error, that is, if the confidence interval of targeted verification minus that policy lies above zero at some grid point of some cell, which would indicate miscalibrated posteriors; and as refuting it if targeted verification never beats the best constant policy in the failure cells. Differences of the order of floating-point rounding, where two policies take the same action on every instance, are treated as ties.

Mass, saving and recovered share. The quantities $m_c(p)$, $\widehat{D}_c(p)$ beside $\widetilde{D}_c(p)$, and $\rho_c(p)$ are reported for every cell, with confidence intervals. They are descriptive and decide no claim.

Sensitivity analyses. The passive threshold is recomputed with the measured cost of leaving a fault in place of $H_F = 0$, with the same monetary values of c_v and $r_K\tau$, so that the sensitivity analysis changes q^* and not the check. The fault rate r_F of Corollary 4 is not measured; the sensitivity analysis keeps $r_F = 0$. Where a cell's scores are close to Gaussian, the closed forms of Proposition 1 may be evaluated as a cross-check, labelled as a calibration model (Remark 2), with the separation also swept directly over a grid; the cross-check decides nothing.

5.8. Results

All results in this subsection are exploratory. The design values are listed in Table 1, the outcome of each criterion in Table 2, and the per-cell counts, costs and quantities in Tables 3–8. Negative outcomes are reported beside the positive one, in the order of Section 5.6.

Interval and critical delay (not evidence). The computed verification intervals agree with the closed forms of Theorem 1 at every grid point, shrink in the delay and are empty exactly at and beyond the critical delay of Theorem 2 (Table 7); since these are theorems, this is a check of the computation, not evidence for them.

Primary exploratory test. On the 6 failure cells of Paper 1, the outcome of the primary test is *supports*, and it is exploratory: the test was specified after Paper 1's cost table had been examined, so it is not a pre-registered confirmation. Every failure cell reaches the minimum agreement share $\bar{\sigma} = 0.9$ over the 528 plausible grid points (smallest share 0.989), no cell is unresolved, and in every failure cell targeted verification beats always leaving, the best constant policy there, at plausible grid points (Table 3). The agreement is not perfect: each failure cell keeps a few disagreeing points (Table 3), and in the contrast regime of V1 many points (341) fall in the band between the two quantile levels, where the prediction depends on a level the claim does not name, and so enter no share. The failure cells are those of Paper 1 on its final basis; they differ from the cells of an earlier, provisional basis on which the same test had weakened the claim, and the cells responsible for that earlier outcome are no longer failure cells, so the present outcome is not

a replication on the same cells. With the measured cost of leaving a fault in place of $H_F = 0$, the outcome is the same (smallest share 0.989); this sensitivity analysis decides nothing.

Policy comparison. The policy comparison, also exploratory, *weakens* the expectation that targeted verification is at least as cheap as every other policy on every cell. Targeted verification is never significantly more costly than always containing, always leaving or verifying everything, in any cell, nor than the passive classifier in any failure cell, and in every failure cell it beats the best constant policy; but in 13 cells where Paper 1’s classifier succeeds it is significantly more costly than the passive classifier at many plausible grid points (Table 4). These differences are not rounding ties, which are treated as ties, and not a failure of the fits, which all converged; where they occur, the realised saving over the passive rule is negative while the saving predicted from the corrected posteriors is positive, which is the sign, stated with the criterion, that the corrected posteriors are miscalibrated in those cells. A cost-ratio threshold, and hence the plug-in saving, is optimal only for calibrated scores [30].

Secondary analyses (decide nothing). The mass in the verification interval is larger in the failure cells than in the success cells at most plausible grid points but not at all of them, and the recovered share is not larger there at most points, so the ordering expected under the replaced formulation of the claim does not hold; nor does its premise that Paper 1’s threshold lies closer to the priors of the failure cells. Both are reported for completeness and decide nothing (Tables 5 and 6; sensitivity analyses in Table 8).

6. Discussion

6.1. What the empirical study shows and does not show

The study of Section 5 does not test the theorems of Section 4, which are proved. What it can test is whether, on the classifier posteriors of a particular set of simulated market designs, the region where targeted verification beats the best constant policy is the region that Theorem 1 predicts from those posteriors. Because the test was specified after Paper 1’s cost table for the same cells had been examined, any outcome of it is exploratory; a confirmation would call for a pre-registered test on fresh data, and a disagreement would point first to the calibration of the corrected posteriors (Section 6.4)

Table 1: Design values of the study, fixed before any quantity of the primary test was computed.

Quantity	Value
Sensitivity s	0.7, 0.8, 0.9, 0.99
Specificity t	0.7, 0.8, 0.9, 0.99
Cost of a check $x = c_v/V_{\max}$	0.001, 0.01, 0.05, 0.1, 0.2, 0.5
Delay $y = r_K\tau/a$	0, 0.01, 0.05, 0.1, 0.25, 0.5
Grid points (plausible / implausible)	576 (528 / 48)
Implausible points	$t = 0.99$ and $x \leq 0.01$
Quantile levels α_{in} , band, α_{out}	0.9, 0.95, 0.99
Confidence level γ	0.95
Agreement threshold $\bar{\sigma} = 1 - 2(1 - \gamma)$	0.9
Bootstrap resamples (within cause)	2000
Passive threshold q^* (Paper 1’s cost matrix, $H_F = 0$)	0.364

Table 2: Outcome of each criterion of Section 5. The primary test is exploratory.

Criterion	Outcome	Exploratory	Decides
Interval and critical delay: closed cross-check passed form against numerical search	no		nothing (a theorem; a mismatch is an error of computation)
Primary exploratory test (Paper 1’s cost matrix)	supports	yes	the outcome of the primary test (exploratory)
Primary test, measured cost of leaving a fault (sensitivity)	supports	yes	nothing (sensitivity)
Policy comparison	weakens	no	the policy-comparison criterion

before it bears on the model. In the failure cells of Paper 1 the outcome of the primary test *supports* the prediction, and it is exploratory; the policy comparison *weakens* the expectation that targeted verification is never beaten, because in several cells where Paper 1’s classifier succeeds the passive classifier is cheaper, which points to miscalibrated corrected posteriors there (Section 5.8). The verification parameters are a grid in model units, not measurements of a particular gateway check, so the study says where a check would pay if it had given characteristics, not whether any existing check has them.

6.2. When the separate channel fails

Every result of Section 4 that involves order flow rests on Assumption 2. If the outcome of the check depends on the order flow given the cause, the verification update is no longer a function of the passive posterior alone and Lemma 2 does not hold. The assumption is a modelling choice about what the check reads, and its fit to a particular verification procedure has to be argued for that procedure.

6.3. One check, then act

The rule is derived for a single check followed by a terminal action (Assumption 3), with the delay charged in full on top of the terminal cost and no order flow used while the check runs (Assumption 4). A fault that costs during the check is covered by Corollary 4, which shortens the critical delay; the fault rate r_F of Corollary 4 is not measured, and the sensitivity analysis keeps $r_F = 0$. Repeated checks, graded responses and evidence that keeps arriving during the delay turn the problem into a sequential one, which this paper does not treat.

6.4. Posteriors and calibration

The saving identity of Lemma 4 holds for the true posterior. Applied to classifier scores it holds only as far as the corrected scores are calibrated, which is why the empirical design reports the plug-in saving $\widehat{D}_c(p)$ beside the realised saving $\widehat{\widehat{D}}_c(p)$ (Section 5.5); a gap between them beyond sampling error points to miscalibration, not to a failure of the lemma.

7. Conclusion

Under the assumption that a check reading authorization is conditionally independent of order flow given the cause, verification is not bound by what order flow can reveal, and it composes with any passive posterior. Transposing the threshold approach, a check pays exactly on an interval of beliefs around the passive containment threshold, and because a true compromise keeps doing harm while the check runs, that interval vanishes at a critical delay in closed form. The interval does not depend on the prior; the saving it brings to a population of cases does, and vanishes as compromise becomes rare.

Section 5 specifies an exploratory test of where targeted verification beats the best constant policy in the cells of Paper 1 where passive classification does no better, with the data, grids, plausible set, policies and outcome criteria fixed before any result. Its outcome supports the prediction that targeted verification beats that policy over the region the two-threshold rule predicts, and it is exploratory, not pre-registered; the policy comparison weakens the expectation that targeted verification is never beaten, since in several cells where passive classification succeeds the passive classifier is cheaper.

Data and code availability

The code that computes the thresholds, the critical delay and the empirical test of Section 5.8, the experiment configurations, the provenance register and the result files from which every number in this article is generated are held in the author’s project repository. They will be deposited in a public archive on acceptance, with a persistent identifier (DOI to be determined). Until then they are available from the corresponding author on reasonable request. The data of the empirical test are the posteriors of the companion study [1], whose deposit they share.

CRedit authorship contribution statement

Chiheb Mejrissi: Conceptualization, Methodology, Formal analysis, Software, Investigation, Validation, Data curation, Visualization, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The author declares no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author used large-language-model assistants (Claude, Anthropic) to draft and revise parts of the text and to write and review parts of the analysis code. After using these tools, the author reviewed, checked and edited the content as needed and takes full responsibility for the content of the published article.

Appendix A. Proofs

This appendix proves the results of Section 4, in the order in which they are stated there.

Proof of Lemma 1. Let $L_c(q) = qC_K + (1 - q)C_F$ (contain) and $L_\ell(q) = qH_K + (1 - q)H_F$ (leave). Their difference is affine:

$$\begin{aligned}\Delta(q) &:= L_\ell(q) - L_c(q) = q(H_K - C_K) - (1 - q)(C_F - H_F) \\ &= qa - (1 - q)b = (a + b)(q - q^*),\end{aligned}$$

with $q^* = b/(a + b)$. By Assumption 1, $a, b > 0$, so $a + b > 0$ and $q^* \in (0, 1)$. Hence $\Delta < 0$ (leave strictly cheaper) for $q < q^*$ and $\Delta > 0$ (contain strictly cheaper) for $q > q^*$. Since $\min\{L_\ell, L_c\} = L_\ell - \max\{0, \Delta\}$,

$$R(q) = L_\ell(q) - (a + b)(q - q^*)^+.$$

R is the minimum of two affine functions, hence continuous and concave; it is piecewise linear with slopes $H_K - H_F$ on $[0, q^*]$ and $C_K - C_F$ on $[q^*, 1]$, which differ by $-(a + b) \neq 0$, so there is exactly one kink, at q^* . $\square$

Proof of Lemma 2. Fix y with $P(Y = y) > 0$ (for densities, replace probabilities of Y by densities; the argument is unchanged), write $q = q(y)$, and let z be an outcome with $P(Z = z | Y = y) > 0$. By Bayes' rule and Assumption 2,

$$P(K | y, z) = \frac{P(z | K) P(y | K) P(K)}{\sum_{c \in \{K, F\}} P(z | c) P(y | c) P(c)}.$$

Divide numerator and denominator by $P(y) = \sum_c P(y | c) P(c)$ and use $P(y | c) P(c) / P(y) = P(c | y)$:

$$P(K | y, z) = \frac{P(z | K) q}{P(z | K) q + P(z | F) (1 - q)}.$$

For $z = +$ this is $qs / (qs + (1 - q)(1 - t)) = q^+$; for $z = -$ it is $q(1 - s) / (q(1 - s) + (1 - q)t) = q^-$. Likewise $P(Z = z | Y = y) = P(z | K) q + P(z | F) (1 - q)$, which is $P(+)$ or $P(-)$ of Definition 2 at $q = q(y)$. So, conditionally on $Y = y$, verification acts on the belief $q(y)$ exactly as in Definition 2 with the same (s, t) , and every quantity built from Definitions 2–3 (R_v , V , S) is evaluated at $q(y)$. Nothing in the argument uses the law of Y beyond $q(y)$; in particular it holds when $P(Y | K) = P(Y | F)$, in which case $q(y)$ equals the prior. Without Assumption 2 the factorisation in the first display fails and the update would depend on y through $P(z | y, c)$. $\square$

Proof of Lemma 3. Step 1 (homogeneous form). For an outcome z write $\alpha_z = P(z | K) q$ and $\beta_z = P(z | F) (1 - q)$, so $P(z) = \alpha_z + \beta_z$ and, by Definition 3, $P(z)R(q^z) = \min\{\alpha_z C_K + \beta_z C_F, \alpha_z H_K + \beta_z H_F\}$. The difference of the two arguments is $(\alpha_z H_K + \beta_z H_F) - (\alpha_z C_K + \beta_z C_F) = \alpha_z a - \beta_z b$, so

$$P(z)R(q^z) = (\alpha_z H_K + \beta_z H_F) - \max\{0, \alpha_z a - \beta_z b\}.$$

Here $\alpha_+ a - \beta_+ b = qsa - (1 - q)(1 - t)b = G^+(q)$ and $\alpha_- a - \beta_- b = q(1 - s)a - (1 - q)tb = -G^-(q)$. Summing over z and using $\alpha_+ + \alpha_- = q$, $\beta_+ + \beta_- = 1 - q$, with $L_\ell(q) = qH_K + (1 - q)H_F$:

$$P(+)R(q^+) + P(-)R(q^-) = L_\ell(q) - \max\{0, G^+(q)\} - \max\{0, -G^-(q)\}.$$

By the proof of Lemma 1, $R(q) = L_\ell(q) - \max\{0, \Delta(q)\}$ with $\Delta(q) = qa - (1 - q)b$, and summing the two outcome terms gives $G^+(q) - G^-(q) = \Delta(q)$. Therefore

$$V(q) = \max\{0, G^+\} + \max\{0, -G^-\} - \max\{0, G^+ - G^-\}. \quad (1)$$

Step 2 (zeros and the value at q^).* G^+ is affine with slope $sa + (1-t)b > 0$ (as $a, b > 0$ and $s > 1-t \geq 0$); G^- is affine with slope $-(tb + (1-s)a) < 0$ (as $t > 1-s \geq 0$). Their unique zeros are q_L and q_U of the statement, and the formulas give $0 \leq q_L$ and $q_U \leq 1$. At q^* , $\Delta(q^*) = 0$, so $G^+(q^*) = G^-(q^*) = q^*sa - (1-q^*)(1-t)b$. Since $q^*a = (1-q^*)b = ab/(a+b)$, this equals $(s+t-1)ab/(a+b)$, which is > 0 by Assumption 3. As G^+ is increasing with $G^+(q^*) > 0$, $q_L < q^*$; as G^- is decreasing with $G^-(q^*) > 0$, $q^* < q_U$. This proves (i).

Step 3 (the tent). Evaluate (1) on three ranges.

- $q \leq q_L$: $G^+ \leq 0$ and (as $q < q_U$) $G^- > 0$, so $G^+ - G^- < 0$; every term of (1) vanishes and $V = 0$.
- $q \geq q_U$: $G^- \leq 0$ and (as $q > q_L$) $G^+ > 0$, so $G^+ - G^- > 0$ and $V = G^+ - G^- - (G^+ - G^-) = 0$.
- $q_L < q < q_U$: $G^+ > 0$ and $G^- > 0$, so $V = G^+ - \max\{0, G^+ - G^-\} = \min\{G^+, G^-\} > 0$.

Hence $V = \max\{0, \min\{G^+, G^-\}\}$ on $[0, 1]$, and $V > 0$ exactly on (q_L, q_U) . On that interval, $G^+ < G^-$ iff $\Delta < 0$ iff $q < q^*$. So $V = G^+$ (affine, strictly increasing) on $[q_L, q^*]$ and $V = G^-$ (affine, strictly decreasing) on $[q^*, q_U]$; V is continuous, and its unique maximiser is q^* , with $V(q^*) = (s+t-1)ab/(a+b) = (s+t-1)aq^*$. This proves (iii) and (iv).

Step 4 (posterior form of (ii)). For $q \in (0, 1)$, $P(+)>0$ and $P(-)>0$ because $s > 0$ and $t > 0$. Using $(a+b)q^* = b$,

$$(a+b)P(+)(q^+ - q^*) = (a+b)qs - bP(+) = qsa - (1-q)(1-t)b = G^+(q),$$

and in the same way $(a+b)P(-)(q^* - q^-) = G^-(q)$. So $G^+(q) > 0 \iff q^+ > q^*$ and $G^-(q) > 0 \iff q^- < q^*$; hence $q \in (q_L, q_U) \iff q^- < q^* < q^+$. At $q \in \{0, 1\}$, $V = 0$, and every defined posterior equals q , so the posteriors cannot straddle $q^* \in (0, 1)$: both sides of (ii) are false. $\square$

Proof of Theorem 1. Assumption 4 is what makes Definition 3 the correct verify-first risk: under K (probability q) verification adds $c_v + r_K\tau$ to the terminal cost, under F it adds c_v , the terminal costs and (s, t) are unchanged by the delay, and the terminal action is chosen optimally on the posterior, giving $c_v + qr_K\tau + \sum_z P(z)R(q^z)$. So $R_v < R$ iff $V > \chi$, with $\chi(q) = c_v + qr_K\tau$ affine and nondecreasing.

Step 1 ($S \subseteq (q_L, q_U)$). By Assumption 3, $\chi(q) \geq 0$ on $[0, 1]$. By Lemma 3, $V = 0$ outside (q_L, q_U) , so $V > \chi$ fails there.

Step 2 (S is an interval). Let $\varphi(q) = \min\{G^+(q), G^-(q)\} - \chi(q)$ on $[0, 1]$. It is the minimum of two affine functions minus an affine function, hence concave and continuous. On (q_L, q_U) , $V = \min\{G^+, G^-\}$ (Lemma 3), and outside it $\min\{G^+, G^-\} \leq 0$ (since $G^+ \leq 0$ for $q \leq q_L$ and $G^- \leq 0$ for $q \geq q_U$), so $\varphi \leq -\chi \leq 0$ there. Hence $S = \{q : \varphi(q) > 0\}$, a strict superlevel set of a continuous concave function: an open (relative to $[0, 1]$) interval, possibly empty. Since $\varphi(0) = \min\{-(1-t)b, tb\} - c_v \leq 0$ and $\varphi(1) = \min\{sa, -(1-s)a\} - c_v - r_K\tau \leq 0$, the endpoints are excluded and S is an open interval in $(0, 1)$.

Step 3 (non-emptiness criterion). On $[q^*, q_U]$, $\varphi = G^- - \chi$ is strictly decreasing (G^- strictly decreasing, χ nondecreasing). On $[q_L, q^*]$, $\varphi = G^+ - \chi$ is affine with slope $\sigma = sa + (1-t)b - r_K\tau$. If $\sigma \leq 0$, φ is nonincreasing on $[q_L, q_U]$, so $\sup_{(q_L, q_U)} \varphi \leq \varphi(q_L) = -\chi(q_L) \leq 0$ and $S = \emptyset$. If $\sigma > 0$, φ is strictly increasing then strictly decreasing, so $\max_{[q_L, q_U]} \varphi = \varphi(q^*) = V(q^*) - \chi(q^*)$. Either way, $S \neq \emptyset$ implies $\varphi(q^*) > 0$ (in the first case S is empty; in the second the maximum is at q^*). Conversely $\varphi(q^*) > 0$ gives $q^* \in S$. Hence $S \neq \emptyset \iff V(q^*) > c_v + q^*r_K\tau$, and then $q^* \in S$. Also, $S \neq \emptyset$ forces $\sigma > 0$: if $\sigma \leq 0$ we showed $S = \emptyset$.

Step 4 (thresholds). Assume $S \neq \emptyset$. Then $\sigma > 0$, and the left endpoint q_1 of S is the zero of the affine function $G^+ - \chi = \sigma q - ((1-t)b + c_v)$ on $[q_L, q^*]$ if $\varphi(q_L) < 0$, or q_L itself if $\varphi(q_L) = 0$; in both cases $q_1 = (c_v + (1-t)b)/\sigma$ (when $\varphi(q_L) = 0$, $\chi(q_L) = 0$, so $c_v = 0$ and $q_L r_K\tau = 0$, and the formula returns q_L). The right endpoint q_2 is the zero of $G^- - \chi = tb - c_v - q(tb + (1-s)a + r_K\tau)$ on $(q^*, q_U]$, i.e. $q_2 = (tb - c_v)/(tb + (1-s)a + r_K\tau)$; its denominator is $\geq tb > 0$. As $G^+(q_1) = \chi(q_1) \geq 0$ we get $q_1 \geq q_L$, and as $G^-(q_2) = \chi(q_2) \geq 0$ we get $q_2 \leq q_U$; $q_1 < q^* < q_2$ because $q^* \in S$ and S is open.

Step 5 (policy). For $q \notin S$ the optimal action is the passive one of Lemma 1; since $q_1 < q^* < q_2$, this is leave on $q \leq q_1$ and contain on $q \geq q_2$. For $q \in S \subseteq (q_L, q_U)$, Lemma 3(ii) gives $q^- < q^* < q^+$, so by Lemma 1 the optimal terminal action after verification is contain on $Z = +$ and leave on $Z = -$.

Step 6 ($c_v = 0, r_K\tau = 0$). Then $\chi \equiv 0$ and $S = \{V > 0\} = (q_L, q_U)$ by Lemma 3(ii); the threshold formulas reduce to q_L and q_U . $\square$

Proof of Theorem 2. Throughout, only τ varies; V, q^*, q_L, q_U do not depend on τ (Assumption 4: s, t and the loss table are delay-free).

(i) *Nesting.* For $q \in [0, 1]$, $\chi_\tau(q) = c_v + qr_K\tau$ is nondecreasing in τ because $qr_K \geq 0$. So $V(q) > \chi_{\tau'}(q)$ implies $V(q) > \chi_\tau(q)$ for $\tau < \tau'$, i.e. $S(\tau') \subseteq S(\tau)$. Now let $r_K > 0$ and $V(q^*) > c_v$. By (ii) below, whose proof does not use (i), $S(\tau) \neq \emptyset$ exactly for $\tau \in [0, \tau^*)$; so on $[0, \tau^*)$ the end points $q_1(\tau), q_2(\tau)$ are given by the formulas of Theorem 1 with positive denominators, and for $\tau \geq \tau^*$ the region is empty and Theorem 1's convention $q_1 = q_2 = q^*$ applies, so no monotonicity is claimed there. On $[0, \tau^*)$: $q_2 = (tb - c_v)/(tb + (1 - s)a + r_K\tau)$ has a positive numerator (because $q_2 > q^* > 0$ and the denominator is positive) and a denominator strictly increasing in τ , so it is strictly decreasing. $q_1 = (c_v + (1 - t)b)/(sa + (1 - t)b - r_K\tau)$ has a positive denominator strictly decreasing in τ and a nonnegative numerator, zero iff $c_v = 0$ and $t = 1$; so q_1 is strictly increasing on $[0, \tau^*)$ unless $c_v = 0$ and $t = 1$, where $q_1 = 0$ on $[0, \tau^*)$.

(ii) *Critical delay.* By Theorem 1, $S(\tau) \neq \emptyset \iff V(q^*) > c_v + q^*r_K\tau$. With $r_K > 0$ and $q^* > 0$, this is $\tau < (V(q^*) - c_v)/(q^*r_K) = \tau^*$, which is > 0 when $V(q^*) > c_v$. By Lemma 3(iv), $V(q^*)/q^* = (s + t - 1)a$, giving the second expression for τ^* .

(iii) *Degenerate cases.* If $V(q^*) \leq c_v$, then $V(q^*) \leq c_v + q^*r_K\tau$ for every $\tau \geq 0$, so $S(\tau) = \emptyset$ by Theorem 1. If $r_K = 0$, χ does not involve τ , so neither does S .

(iv) *How the interval vanishes.* By hypothesis $r_K > 0$ and $V(q^*) > c_v$, so τ^* is defined by (ii); let $\tau \uparrow \tau^*$. The denominator of q_2 is continuous and positive at τ^* , so $q_2(\tau) \rightarrow q_2(\tau^*)$, the zero of $G^- - \chi_{\tau^*}$. At τ^* , $G^-(q^*) = V(q^*) = \chi_{\tau^*}(q^*)$, so that zero is q^* (it is unique: $G^- - \chi_{\tau^*}$ is strictly decreasing). For q_1 : at τ^* the left-piece function $G^+ - \chi_{\tau^*}$ vanishes at q^* and equals $-\chi_{\tau^*}(q_L) = -(c_v + q_L r_K \tau^*)$ at q_L . If $c_v > 0$ or $t < 1$ (the latter giving $q_L > 0$), this is < 0 , so the slope $\sigma(\tau^*) = sa + (1 - t)b - r_K\tau^*$ is > 0 , the formula for q_1 is continuous at τ^* , and $q_1(\tau) \rightarrow$ the zero of $G^+ - \chi_{\tau^*}$, which is q^* . If $c_v = 0$ and $t = 1$, then $q_L = 0$, $G^+(q) = qsa$ and $\chi_\tau(q) = qr_K\tau$, and $\tau^* = sa/r_K$ (as $s + t - 1 = s$). For $\tau < \tau^*$, $G^+ - \chi_\tau = q(sa - r_K\tau) > 0$ on $(0, q^*]$, so $S(\tau) = (0, q_2(\tau))$ with $q_2(\tau) \downarrow q^*$ by (i) and the limit above; at τ^* , $G^+ - \chi_{\tau^*} \equiv 0$ on $[0, q^*]$ and $S(\tau^*) = \emptyset$ by (ii). So the interval drops from $(0, q^*)$ (in the limit) to $\emptyset$. $\square$

Proof of Corollary 1. Reduction to a constant belief. By Assumption 2 the joint law of (M, Y, Z) is $P(M = m) P_m^Y(dy) P(Z = z | m)$. Let $\mu = P_K^Y = P_F^Y$. Then Y has law μ and is independent of (M, Z) , and $P(M = m, Z = z) = P(m)P(z | m)$. Write $P(z, K) = P(z | K)p$ and $P(z, F) = P(z |$

$F)(1-p)$ as in Definition 3. Conditioning on $Y = y$ in Definition 4 gives $J(\delta, \pi) = \int \ell(y) \mu(dy)$ with

$$\begin{aligned} \ell(y) = & (1 - \delta(y)) \left[\pi_0(y) (pC_K + (1-p)C_F) \right. \\ & \left. + (1 - \pi_0(y)) (pH_K + (1-p)H_F) \right] \\ & + \delta(y) \left[c_v + pr_K\tau + \sum_{z \in \{+, -\}} \left(\pi_z(y) (P(z, K)C_K + P(z, F)C_F) \right. \right. \\ & \left. \left. + (1 - \pi_z(y)) (P(z, K)H_K + P(z, F)H_F) \right) \right]. \end{aligned}$$

Here c_v and $r_K\tau$ are paid only on verification, the latter only under K (Assumption 4). Each expression of the form $\pi x + (1 - \pi) x'$ with $\pi \in [0, 1]$ is at least $\min\{x, x'\}$. So the first bracket is at least $R(p)$ (Definition 1). The second is at least $c_v + pr_K\tau + \sum_z \min\{P(z, K)C_K + P(z, F)C_F, P(z, K)H_K + P(z, F)H_F\} = R_v(p)$, by the reading of Definition 3. Hence, for every y ,

$$\ell(y) \geq (1 - \delta(y))R(p) + \delta(y)R_v(p) \geq \min\{R(p), R_v(p)\},$$

and $J \geq \min\{R(p), R_v(p)\}$. The constant policy that verifies iff $R_v(p) < R(p)$ and takes the minimizing terminal action in each case attains equality. So

$$R_{Y,v}^* = \min\{R(p), R_v(p)\}, \quad R_Y^* = R(p)$$

(the second with $\delta \equiv 0$). The attaining policy depends on p only, not on y . When $p \in S$ it is the verify branch of Theorem 1 at $q = p$: contain on $Z = +$ and leave on $Z = -$, because $p^- < q^* < p^+$ by Lemma 3(ii). Otherwise it is the passive rule of Lemma 1.

(i) By Assumption 1, $\min\{C_K, H_K\} = C_K$ and $\min\{C_F, H_F\} = H_F$, so $R_{\text{oracle}} = pC_K + (1-p)H_F$. Subtracting it from both terms of $R(p)$ gives $\text{CoA}^* = \min\{(1-p)(C_F - H_F), p(H_K - C_K)\} = \min\{pa, (1-p)b\}$. This is > 0 since $a, b > 0$ and $0 < p < 1$.

(ii) By Definition 3, $R_v(p) = \chi(p) + R(p) - V(p)$. So

$$\text{CoA}_v^* = \min\{R(p), R_v(p)\} - R_{\text{oracle}} = \text{CoA}^* - (V(p) - \chi(p))^+.$$

With $G^\pm$ as in Lemma 3,

$$\begin{aligned} \text{CoA}^* - E(p) &= \min\{pa - p(1-s)a - (1-p)(1-t)b, \\ &\quad (1-p)b - p(1-s)a - (1-p)(1-t)b\} - \chi(p) \\ &= \min\{G^+(p), G^-(p)\} - \chi(p). \end{aligned}$$

By Lemma 3, $V = \max\{0, \min\{G^+, G^-\}\}$, and $\chi(p) \geq 0$ by Assumption 3. Suppose $\min\{G^+, G^-\} > \chi$. Then $V = \min\{G^+, G^-\}$ and $(V - \chi)^+ = \min\{G^+, G^-\} - \chi$. Otherwise $V \leq \max\{0, \chi\} = \chi$, and $(V - \chi)^+ = 0 = (\min\{G^+, G^-\} - \chi)^+$. In both cases $(V - \chi)^+ = (\text{CoA}^* - E)^+$, so $\text{CoA}_v^* = \text{CoA}^* - (\text{CoA}^* - E)^+ = \min\{\text{CoA}^*, E(p)\}$. (Direct check: $E(p)$ is the cost of the verify branch above minus R_{oracle} . That is c_v , plus $r_K\tau$ under K , a miss a with probability $1 - s$ under K , and a false containment b with probability $1 - t$ under F .)

(iii) By (ii), $\text{CoA}_v^* < \text{CoA}^* \iff V(p) > \chi(p) \iff p \in S$ (Definition 3). By Theorem 1, $S \neq \emptyset \iff V(q^*) > c_v + q^*r_K\tau$, and then $S = (q_1, q_2)$. If $r_K > 0$ and $V(q^*) > c_v$, Theorem 2(ii) turns the non-emptiness condition into $\tau < \tau^*$. If $V(q^*) \leq c_v$, then $S(\tau) = \emptyset$ for all $\tau \geq 0$ by Theorem 2(iii).

(iv) The reduction is $g(p) = (V(p) - \chi(p))^+$, and $V - \chi \leq -\chi \leq 0$ outside (q_L, q_U) . On $[q^*, q_U]$, $V - \chi = G^- - \chi$ is strictly decreasing (Lemma 3(iii); χ is nondecreasing). On $[q_L, q^*]$, $V - \chi$ is affine. It is ≤ 0 at q_L and equals $V(q^*) - \chi(q^*)$ at q^* . So its maximum over $[q_L, q^*]$ is attained at an endpoint, and $\max_p g(p) = (V(q^*) - \chi(q^*))^+ = (V(q^*) - c_v - q^*r_K\tau)^+$, attained at q^* . When this maximum is positive, g is strictly increasing on $[q_L, q^*]$ (an affine function going from ≤ 0 to a positive value) wherever it is positive, and strictly decreasing on $[q^*, q_U]$. So the maximizer is unique.

(v) Since $\text{CoA}^* > 0$ by (i), $\text{CoA}_v^* = 0$ iff $E(p) = 0$. Each of the four terms of $E(p)$ is nonnegative (Assumption 3). With $p \in (0, 1)$ and $a, b > 0$, all four vanish iff $c_v = 0$, $r_K\tau = 0$, $s = 1$ and $t = 1$. $\square$

Proof of Corollary 2. Fix order-flow laws P_K^Y, P_F^Y and $p \in (0, 1)$. By Assumption 1, $\min\{C_K, H_K\} = C_K$ and $\min\{C_F, H_F\} = H_F$, so $R_{\text{oracle}} = pC_K + (1 - p)H_F$.

(i) *Verify-everything.* With $\delta \equiv 1$, $\pi_+ \equiv 1$, $\pi_- \equiv 0$, the integrand of J in Definition 4 does not depend on Y : the operator pays c_v , pays $r_K\tau$ under K (Assumption 4), contains on $Z = +$ and leaves on $Z = -$. By Assumption 2, Z given M has the law of Definition 2 whatever Y , so

$$J = c_v + pr_K\tau + p(sC_K + (1 - s)H_K) + (1 - p)((1 - t)C_F + tH_F).$$

Subtracting R_{oracle} and using $H_K - C_K = a$, $C_F - H_F = b$,

$$J - R_{\text{oracle}} = c_v + pr_K\tau + p(1 - s)a + (1 - p)(1 - t)b = E(p).$$

(ii) *Bounds.* The constant passive policies $\pi_0 \equiv 1$ (always contain) and $\pi_0 \equiv 0$ (always leave) cost $pC_K + (1 - p)C_F$ and $pH_K + (1 - p)H_F$, that is

$R_{\text{oracle}} + (1-p)b$ and $R_{\text{oracle}} + pa$. So $R_Y^* \leq R_{\text{oracle}} + \min\{pa, (1-p)b\}$, which is the first bound. Every passive policy is a verify-first policy ($\delta \equiv 0$), so $R_{Y,v}^* \leq R_Y^*$; by (i), $R_{Y,v}^* \leq R_{\text{oracle}} + E(p)$. Hence $\text{CoA}_v^* \leq \min\{\text{CoA}^*, E(p)\}$, and the first bound gives the second inequality.

(iii) *Equality in the confounded case.* If $P_K^Y = P_F^Y$, Corollary 1(i) gives $\text{CoA}^* = \min\{pa, (1-p)b\}$ and Corollary 1(ii) gives $\text{CoA}_v^* = \min\{\text{CoA}^*, E(p)\}$, so every inequality in (ii) is an equality. Together with (ii), which holds for every pair of order-flow laws, the suprema over pairs of order-flow laws of CoA^* and of CoA_v^* are $\min\{pa, (1-p)b\}$ and $\min\{pa, (1-p)b, E(p)\}$, and both are attained by any confounded pair (e.g. $P_K^Y = P_F^Y$ a point mass).

(iv) *Gap.* Write $m = \min\{pa, (1-p)b\}$. The gap between the two suprema is $m - \min\{m, E(p)\} = (m - E(p))^+$. In the confounded case $m = \text{CoA}^*$ and Corollary 1(ii) gives $\min\{m, E(p)\} = m - (V(p) - \chi(p))^+$, so the gap is $(V(p) - \chi(p))^+$. It is positive if and only if $V(p) > \chi(p)$, that is (Definition 3) if and only if $p \in S$. $\square$

Proof of Corollary 3. Fix $q \in [0, 1]$, $\tau \geq 0$ and $\theta \in [0, 1]$. Assumption 3 gives $s + t > 1$ with $s, t \leq 1$, so $s > 0$ and $t > 0$; hence $\sigma_\theta \in [s, 1] \subset (0, 1]$.

Four plans. Each minimum in $R_v^\theta(q)$ is over the two terminal actions after outcome z , so $R_v^\theta(q)$ is the least of the costs of the four plans (action after $Z = +$, action after $Z = -$); randomizing between actions cannot do better, since the cost is linear in the probabilities. Write $L = qH_K + (1-q)H_F$ and $C = qC_K + (1-q)C_F$, so $R(q) = \min\{L, C\}$, and let $G^\pm$ be as in Lemma 3. For each plan, the saving $R(q) - \text{cost}$ is:

- *contain, contain:* cost $c_v + C + qr_K\tau$, saving $\leq -c_v - qr_K\tau \leq 0$;
- *leave, leave:* cost $c_v + L + \theta qr_K\tau$, saving ≤ 0 ;
- *leave after +, contain after -:* the terminal part is $T' = P(-, K)C_K + P(-, F)C_F + P(+, K)H_K + P(+, F)H_F$, with $L - T' = q(1-s)a - (1-q)tb = -G^-(q)$ and $C - T' = -qsa + (1-q)(1-t)b = -G^+(q)$; the saving is at most $\min\{-G^+(q), -G^-(q)\}$. Now $G^+(q) = q(sa + (1-t)b) - (1-t)b$ is increasing and vanishes at q_L , and $G^-(q) = tb - q(tb + (1-s)a)$ is decreasing and vanishes at q_U (both slopes are non-zero since $a, b, s, t > 0$); as $q_L < q_U$ (Lemma 3(i)), at every q one of $G^+(q) \geq 0$, $G^-(q) \geq 0$ holds, so this saving is ≤ 0 ;
- *contain after +, leave after -:* the terminal part is $T = P(+, K)C_K + P(+, F)C_F + P(-, K)H_K + P(-, F)H_F$, with $L - T = qsa - (1 -$

$q)(1-t)b = G^+(q)$ and $C - T = (1-q)tb - q(1-s)a = G^-(q)$; the delay part is $P(+, K) r_K \tau + \theta P(-, K) r_K \tau = \sigma_\theta q r_K \tau$. The saving is $\min\{G^+(q), G^-(q)\} - c_v - \sigma_\theta q r_K \tau$.

(i) $q \in S_\theta(\tau)$ if and only if some plan has a strictly positive saving. The first three never do, so $q \in S_\theta(\tau)$ if and only if $\min\{G^+(q), G^-(q)\} > c_v + \sigma_\theta q r_K \tau$, and then the fourth plan is the unique optimal one. The right-hand side is ≥ 0 ($c_v \geq 0$, $r_K \geq 0$, Assumption 3), so by Lemma 3 ($V = \max\{0, \min\{G^+, G^-\}\}$) the condition is $V(q) > c_v + \sigma_\theta q r_K \tau$. This is the condition $V(q) > \chi(q)$ of Definition 3 with r_K replaced by $r'_K = \sigma_\theta r_K \geq 0$. The proofs of Theorems 1 and 2 use Assumption 4 only to write the verification cost as $c_v + q r_K \tau$ with $r_K \geq 0$, and the optimal post-verification action there is the fourth plan, as here; so both theorems hold under Assumption 4 $_\theta$ with r'_K in place of r_K .

(ii) σ_θ is nondecreasing in θ (its derivative is $1-s \geq 0$), with $\sigma_1 = 1$. For fixed q and τ , $c_v + \sigma_\theta q r_K \tau$ is nondecreasing in θ because $q r_K \tau \geq 0$; by (i), $S_{\theta'}(\tau) \subseteq S_\theta(\tau)$ for $\theta \leq \theta'$. At $\theta = 1$ the condition is that of Definition 3, so $S_1(\tau) = S(\tau)$.

(iii) Let $r_K > 0$ and $V(q^*) > c_v$. Then $r'_K = \sigma_\theta r_K > 0$, and Theorem 2(ii) with r'_K (valid by (i)) gives $S_\theta(\tau) \neq \emptyset$ if and only if $\tau < (V(q^*) - c_v)/(q^* \sigma_\theta r_K) = \tau^*/\sigma_\theta$. Since $s \leq \sigma_\theta \leq 1$, $\tau^* \leq \tau^*/\sigma_\theta \leq \tau^*/s$, and $\tau_\theta^* = \tau^*$ if and only if $\sigma_\theta = 1$, that is $(1-\theta)(1-s) = 0$: $\theta = 1$ or $s = 1$. $\square$

Proof of Corollary 4. Verify-first risk. Under Assumption 4 $_F$, verifying adds $c_v + r_K \tau$ to the terminal cost under K and $c_v + r_F \tau$ under F ; neither depends on the terminal action, so the optimal action after each outcome is the one of Definition 3, and the verify-first risk is $c_v + q r_K \tau + (1-q) r_F \tau + P(+, K) R(q^+) + P(-, K) R(q^-) = R_v(q) + (1-q) r_F \tau$. Hence verifying is strictly cheaper iff $V(q) > \chi_F(q)$. χ_F is affine and $\chi_F \geq 0$ on $[0, 1]$, since $\chi_F(0) = c_v + r_F \tau \geq 0$ and $\chi_F(1) = c_v + r_K \tau \geq 0$ (Assumption 3, $r_F \geq 0$).

(i) As in Steps 1–2 of the proof of Theorem 1, which use only that the verification cost is affine and ≥ 0 : $S \subseteq (q_L, q_U)$, and $S = \{q : \varphi(q) > 0\}$ with $\varphi = \min\{G^+, G^-\} - \chi_F$ concave and continuous, so S is an open interval in $(0, 1)$, possibly empty. On $[q_L, q^*]$, $\varphi = G^+ - \chi_F$ is affine with slope $\sigma_1 = sa + (1-t)b - (r_K - r_F)\tau$; on $[q^*, q_U]$, $\varphi = G^- - \chi_F$ is affine with slope $-\sigma_2$, $\sigma_2 = tb + (1-s)a + (r_K - r_F)\tau$. At the ends, $\varphi(q_L) = -\chi_F(q_L) \leq 0$ and $\varphi(q_U) = -\chi_F(q_U) \leq 0$. If $\sigma_1 \leq 0$, then φ is nonincreasing on $[q_L, q^*]$ and, being concave, also on $[q^*, q_U]$, so $\varphi \leq \varphi(q_L) \leq 0$ on $[q_L, q_U]$ and $S = \emptyset$. If $\sigma_2 \leq 0$, symmetrically φ is nondecreasing on $[q_L, q_U]$, $\varphi \leq \varphi(q_U) \leq 0$ and

$S = \emptyset$. If $\sigma_1, \sigma_2 > 0$, φ is maximal at q^* . So in every case $S \neq \emptyset$ iff $\varphi(q^*) > 0$, i.e. iff $V(q^*) > \chi_F(q^*) = c_v + \bar{r}\tau$, and then $\sigma_1, \sigma_2 > 0$ and $q^* \in S$. The left end q_1 is the unique zero of $G^+ - \chi_F = \sigma_1 q - ((1-t)b + c_v + r_F\tau)$ in $[q_L, q^*)$ (it lies there because $\varphi(q_L) \leq 0 < \varphi(q^*)$), which is the stated formula; the right end q_2 is the unique zero of $G^- - \chi_F = tb - c_v - r_F\tau - \sigma_2 q$ in $(q^*, q_U]$, the stated formula. $q_1 < q^* < q_2$ because $q^* \in S$ and S is open. The policy follows as in Step 5 of the proof of Theorem 1: outside S the passive rule of Lemma 1; on $S \subseteq (q_L, q_U)$, Lemma 3(ii) gives $q^- < q^* < q^+$, so contain on $Z = +$ and leave on $Z = -$.

(ii) For fixed q , $\chi_F(q) = c_v + \tau(qr_K + (1-q)r_F)$ is nondecreasing in τ , so $V(q) > \chi_F(q)$ at τ' implies it at $\tau < \tau'$: $S(\tau') \subseteq S(\tau)$. By (i), $S(\tau) \neq \emptyset$ iff $V(q^*) - c_v > \bar{r}\tau$. If $V(q^*) \leq c_v$ this fails for every $\tau \geq 0$ ($\bar{r} \geq 0$). If $\bar{r} > 0$ and $V(q^*) > c_v$, it is $\tau < (V(q^*) - c_v)/\bar{r} = \tau_F^*$.

(iii) If $r_K > 0$, then $\tau^* = (V(q^*) - c_v)/(q^*r_K)$ (Theorem 2(ii)), so $\tau_F^* = \tau^* q^* r_K / \bar{r}$. Since $\bar{r} - q^* r_K = (1 - q^*)r_F \geq 0$ with $1 - q^* > 0$ (Lemma 1), $\tau_F^* \leq \tau^*$, with equality iff $r_F = 0$. $\square$

Proof of Remark 1. Step 1 of the proof of Lemma 3 used only Definitions 1–3 and holds without Assumptions 1 and 3: $V = \max\{0, G^+\} + \max\{0, -G^-\} - \max\{0, G^+ - G^-\}$, with $G^+ - G^- = \Delta = qa - (1-q)b$.

(a) If $s + t = 1$, then $P(z | K) = P(z | F)$ for both outcomes, so $q^z = q$ whenever $P(z) > 0$. Then $\sum_z P(z)R(q^z) = R(q)$ (the $P(z) = 0$ term is zero by Definition 3), so $V \equiv 0$. As $\chi \geq 0$, $V > \chi$ never holds and $S = \emptyset$.

(b1) If $a \leq 0 \leq b$, then $\Delta(q) = qa - (1-q)b \leq 0$ (the difference of the proof of Lemma 1) on $[0, 1]$, so leaving is weakly optimal everywhere and $R = L_\ell$ is affine. Each outcome term satisfies $\alpha_z a - \beta_z b \leq 0$ (as $\alpha_z, \beta_z \geq 0$), so $\sum_z P(z)R(q^z) = \sum_z (\alpha_z H_K + \beta_z H_F) = L_\ell(q) = R(q)$ and $V \equiv 0$. The case $b \leq 0 \leq a$ is symmetric ($\Delta \geq 0$, contain weakly optimal everywhere, each $\alpha_z a - \beta_z b \geq 0$, so $\sum_z P(z)R(q^z) = L_c(q) = R(q)$). In both, $S = \emptyset$ since $\chi \geq 0$. These two sign patterns cover every failure of Assumption 1 except $a < 0$ and $b < 0$: if one of a, b is ≤ 0 and the other is ≥ 0 , one of the two patterns holds, and the only remaining way to have $a \leq 0$ or $b \leq 0$ is $a < 0$ and $b < 0$.

(b2) Let $a < 0$ and $b < 0$. Then $a + b < 0$ and $q^* = b/(a + b) \in (0, 1)$, and $\Delta(q) = (a + b)(q - q^*)$ is > 0 (contain strictly cheaper) for $q < q^*$ and < 0 for $q > q^*$. Exchange the names of the two terminal actions: the new costs are $C'_K = H_K$, $C'_F = H_F$, $H'_K = C_K$, $H'_F = C_F$, so $a' = H'_K - C'_K = -a > 0$, $b' = C'_F - H'_F = -b > 0$ and $b'/(a' + b') = q^*$: the renamed problem

satisfies Assumption 1. R is the minimum over the two actions (Definition 1) and each term $P(z)R(q^z)$ is the minimum of the two joint costs (Definition 3), so R , V and S do not depend on the names; χ involves no terminal action; Assumptions 2–4 do not refer to the names (Assumption 4 charges $r_K\tau$ whichever terminal action is taken). In Definition 4, renaming maps a verify-first policy $(\delta, \pi_0, \pi_+, \pi_-)$ to $(\delta, 1 - \pi_0, 1 - \pi_+, 1 - \pi_-)$, a bijection that preserves J , and R_{oracle} is symmetric in the two actions. Hence every result stated under Assumption 1 holds for the renamed problem with $(a', b') = (-a, -b)$; read back in the original names, contain and leave are exchanged in each policy, which gives the rule stated in (b2).

(c) Take $c_v < 0$, $t < 1$ (so $q_L > 0$ under Assumption 1), $r_K > 0$, and $\tau > 0$ small enough that $V(q^*) > q^*r_K\tau$. Choose $c_v \in (-q_Lr_K\tau, 0)$. Then $q^* \in S$, because $V(q^*) > q^*r_K\tau > \chi(q^*)$. At $q = 0$, $V(0) = 0 > c_v = \chi(0)$, so $0 \in S$. At $q = q_L$, $V(q_L) = 0 < c_v + q_Lr_K\tau = \chi(q_L)$, so $q_L \notin S$. Since $0 < q_L < q^*$, S is not an interval. $\square$

Proof of Lemma 4. Write $f_K = f_K^c$, $f_F = f_F^c$, $f_p = pf_K + (1 - p)f_F$ (the ν -density of P_p), $q = q_p$, and $g = (V - \chi)^+$. Let $N = \{y : f_p(y) = 0\}$; then $P_p(N) = 0$, and q is a measurable map from N^c to $[0, 1]$.

Properties of g . By Lemma 3, $V = \max\{0, \min\{G^+, G^-\}\}$ with $G^\pm$ affine, so V is continuous on $[0, 1]$ and $0 \leq V \leq V(q^*)$; $\chi(q) = c_v + qr_K\tau$ is affine and ≥ 0 (Assumption 3). Hence g is continuous, $0 \leq g \leq V \leq V(q^*)$, and $g > 0$ exactly on S (Definition 3). At $q = 0$: $G^+(0) = -(1 - t)b \leq 0$, so $V(0) = 0$ and $g(0) = 0$.

(i) *Saving.* Fix a verify-first policy $(\delta, \pi_0, \pi_+, \pi_-)$. By Assumption 2 the joint law of (M, Y, Z) has density $P(M = m) f_m(y) P(Z = z | m)$ with respect to (counting $\times \nu \times$ counting). Integrating out m and z in Definition 4,

$$J(\delta, \pi) = \int_{N^c} f_p(y) \ell_{q(y)}(y) \nu(dy),$$

where $\ell_q(y)$ is the expression $\ell(y)$ of the proof of Corollary 1 with p replaced by q , since $pf_K(y)/f_p(y) = q(y)$ and $(1 - p)f_F(y)/f_p(y) = 1 - q(y)$. The pointwise argument of that proof gives $\ell_{q(y)}(y) \geq (1 - \delta(y)) R(q(y)) + \delta(y) R_v(q(y)) \geq \min\{R, R_v\}(q(y))$, and, for passive policies ($\delta \equiv 0$), $\ell_{q(y)}(y) \geq R(q(y))$. All costs are bounded, so every integral is finite. Equality holds for the following measurable policy: $\delta(y) = \mathbb{1}[q(y) \in S]$ (S is an interval, Theorem 1); $\pi_0(y) = \mathbb{1}[q(y) > q^*]$ (Lemma 1); $\pi_+ \equiv 1$, $\pi_- \equiv 0$. The last two are optimal on $\{q(y) \in S\}$: there $V(q) > \chi(q) \geq 0$, so $G^+(q) > 0$ and $G^-(q) > 0$

(Lemma 3). Since $G^+(q) = P(+, K) a - P(+, F) b$, containing is strictly better than leaving after $Z = +$. Since $G^-(q) = P(-, F) b - P(-, K) a$, leaving is strictly better after $Z = -$. With $\delta \equiv 0$ and the same π_0 , equality holds in the passive bound. Hence

$$R_{Y,v}^* = \mathbb{E}_p[\min\{R, R_v\}(q(Y))], \quad R_Y^* = \mathbb{E}_p[R(q(Y))].$$

By Definition 3, $R - R_v = V - \chi$, so $R - \min\{R, R_v\} = (V - \chi)^+ = g$, and $R_Y^* - R_{Y,v}^* = \mathbb{E}_p[g(q(Y))] = D_c(p)$. Subtracting R_{oracle} from both infima gives $D_c(p) = \text{CoA}_c^*(p) - \text{CoA}_{v,c}^*(p)$. For the bounds: $D_c \geq 0$ since $g \geq 0$. $\text{CoA}_{v,c}^* \geq 0$ because, for every policy, the integrand of J is at least $\min\{L(\text{contain}, M), L(\text{leave}, M)\}$ ($c_v \geq 0$, $r_K \tau \geq 0$), and the expectation of that is R_{oracle} ; so $D_c \leq \text{CoA}_c^*$. Finally $g \leq V(q^*) \mathbb{1}_S$ gives $D_c(p) \leq V(q^*) m_c(p)$.

Pointwise limit under F . Let y satisfy $f_F(y) > 0$ (a set of full P_F^c -measure). Then $y \notin N$ for every $p \in (0, 1)$, and

$$0 \leq q_p(y) \leq \frac{p f_K(y)}{(1-p) f_F(y)} \longrightarrow 0 \quad (p \rightarrow 0),$$

because $f_K(y) < \infty$.

(ii) Split $D_c(p) = p \mathbb{E}_K[g(q_p(Y))] + (1-p) \mathbb{E}_F[g(q_p(Y))]$, where $\mathbb{E}_m$ is the expectation under P_m^c (the event N is null under both). The first term is at most $p V(q^*) \rightarrow 0$. In the second, $g(q_p(y)) \rightarrow g(0) = 0$ for P_F^c -almost every y , by the pointwise limit and continuity of g . The integrand is bounded by the constant $V(q^*)$, so dominated convergence gives $\mathbb{E}_F[g(q_p(Y))] \rightarrow 0$. Hence $D_c(p) \rightarrow 0$.

(iii) If $S = \emptyset$, then $m_c \equiv 0$. Let $S = (q_1, q_2) \neq \emptyset$. By Theorem 1 the denominator of q_1 is positive, so $q_1 > 0$ iff $c_v + (1-t)b > 0$, i.e. iff $c_v > 0$ or $t < 1$ ($b > 0$, Assumption 1). Suppose $q_1 > 0$. Then $m_c(p) \leq p + (1-p) P_F^c(q_p(Y) > q_1)$. For P_F^c -almost every y , $q_p(y) \rightarrow 0 < q_1$, so $\mathbb{1}_{[q_p(y) > q_1]} \rightarrow 0$. Dominated convergence (bound 1) gives $P_F^c(q_p(Y) > q_1) \rightarrow 0$, hence $m_c(p) \rightarrow 0$.

(iv) Let $S \neq \emptyset$, $c_v = 0$ and $t = 1$. Then $q_1 = 0$ (Theorem 1) and $S = (0, q_2)$ with $q_2 > q^* > 0$. Write $m_c(p) = p P_K^c(q_p(Y) \in S) + (1-p) P_F^c(q_p(Y) \in S)$; the first term is at most $p \rightarrow 0$. Take y with $f_F(y) > 0$. If $f_K(y) = 0$, then $q_p(y) = 0 \notin S$ for every p . If $f_K(y) > 0$, then $q_p(y) > 0$, and $q_p(y) < q_2$ for all small enough p by the pointwise limit, so $q_p(y) \in S$ eventually. Thus

$\mathbb{K}[q_p(y) \in S] \rightarrow \mathbb{K}[f_K(y) > 0]$ for P_F^c -almost every y . Dominated convergence gives $P_F^c(q_p(Y) \in S) \rightarrow P_F^c(f_K(Y) > 0)$, and $(1 - p) \rightarrow 1$.

When the limit is positive. Let $A = \{y : f_K(y) > 0\}$, so $P_K^c(A) = 1$. If $P_F^c(A) = 0$, then A separates the two laws and they are mutually singular. Conversely, if $P_K^c(B) = 1$ and $P_F^c(B) = 0$ for some measurable B , then $\int_{B^c} f_K d\nu = P_K^c(B^c) = 0$, so $f_K = 0$ ν -almost everywhere on B^c ; since $P_F^c \ll \nu$, $P_F^c(A \cap B^c) = 0$, and $P_F^c(A) \leq P_F^c(B) + P_F^c(A \cap B^c) = 0$. Hence the limit $P_F^c(A)$ is zero if and only if P_K^c and P_F^c are mutually singular. That the saving still vanishes is (ii). $\square$

Proof of Proposition 1. Law of the log-likelihood ratio. Let $\Delta = \mu_K - \mu_F$ and $\bar{\mu} = (\mu_K + \mu_F)/2$. The Gaussian densities are positive everywhere, and

$$\Lambda(y) = \log \frac{f_K(y)}{f_F(y)} = \Delta^\top \Sigma_Y^{-1}(y - \bar{\mu}).$$

Under $M = K$, $y - \bar{\mu} \sim \mathcal{N}(\Delta/2, \Sigma_Y)$, so $\Lambda \sim \mathcal{N}(d^2/2, d^2)$. Under $M = F$, $\Lambda \sim \mathcal{N}(-d^2/2, d^2)$. By Definition 5, $\text{logit } q_p(Y) = \ell + \Lambda(Y)$, which is strictly increasing in Λ . For $-\infty \leq u < v \leq \infty$, with Λ continuous ($d > 0$),

$$P_K^c(\text{logit } q_p(Y) \in (u, v)) = P\left(\frac{d^2}{2} + dW \in (u - \ell, v - \ell)\right) = \Pi_K(u, v),$$

with $W \sim \mathcal{N}(0, 1)$, and likewise $P_F^c(\cdot) = \Pi_F(u, v)$. Open and half-open intervals have the same probability.

Mass. $q_p(Y) \in S = (q_1, q_2)$ iff $\text{logit } q_p(Y) \in (\ell_1, \ell_2)$ (with the stated conventions when $q_1 = 0$ or $q_2 = 1$; $q_p(Y) \in (0, 1)$ always). Since $P_p = pP_K^c + (1 - p)P_F^c$, this gives m_c .

Saving. On S , $V > \chi \geq 0$, so $V = \min\{G^+, G^-\}$ (Lemma 3). $G^+ - G^- = qa - (1 - q)b$ is ≤ 0 iff $q \leq q^*$. So on $(q_1, q^*]$, $g := (V - \chi)^+ = G^+ - \chi = \alpha_1 q - \beta_1$; on $[q^*, q_2)$, $g = G^- - \chi = \beta_2 - \alpha_2 q$. Outside S , $g = 0$. Expanding $G^\pm$ from Lemma 3 and $\chi = c_v + qr_K\tau$ gives the stated α_i, β_i . Theorem 1 gives $q_1 = \beta_1/\alpha_1$ and $q_2 = \beta_2/\alpha_2$. For any Borel $A \subseteq (0, 1)$,

$$\mathbb{E}_p[q_p(Y) \mathbb{K}_A(q_p(Y))] = \int \frac{pf_K}{f_p} \mathbb{K}_A(q_p) f_p d\nu = p P_K^c(q_p(Y) \in A).$$

Take $A_1 = (q_1, q^*]$ and $A_2 = (q^*, q_2)$, and write $P_p = pP_K^c + (1 - p)P_F^c$. Then

$$\mathbb{E}_p[(\alpha_1 q_p - \beta_1) \mathbb{K}_{A_1}(q_p)] = p(\alpha_1 - \beta_1) P_K^c(A_1) - (1 - p) \beta_1 P_F^c(A_1),$$

$$\mathbb{E}_p[(\beta_2 - \alpha_2 q_p) \mathbb{K}_{A_2}(q_p)] = p(\beta_2 - \alpha_2) P_K^c(A_2) + (1 - p) \beta_2 P_F^c(A_2),$$

where $P_m^c(A)$ abbreviates $P_m^c(q_p(Y) \in A)$. Substituting $P_m^c(A_1) = \Pi_m(\ell_1, \ell^*)$ and $P_m^c(A_2) = \Pi_m(\ell^*, \ell_2)$, and summing, gives D_c .

Confounded limit $d \rightarrow 0$. For fixed finite $u < v$ and $\ell \notin \{u, v\}$, $(v - \ell)/d \mp d/2 \rightarrow \pm\infty$ according to the sign of $v - \ell$. So $\Pi_K(u, v), \Pi_F(u, v) \rightarrow \mathbb{K}[u < \ell < v]$, and the same holds with an infinite endpoint. At $\ell = u$ or $\ell = v$ the corresponding Φ term tends to $\Phi(0) = 1/2$. Hence $m_c(p) \rightarrow \mathbb{K}[\ell_1 < \ell < \ell_2] = \mathbb{K}[p \in S]$ for $p \notin \{q_1, q_2\}$. For D_c , take $p \in (q_1, q^*)$: the limit is $p(\alpha_1 - \beta_1) - (1 - p)\beta_1 = \alpha_1 p - \beta_1 = g(p)$. Take $p \in (q^*, q_2)$: the limit is $\beta_2 - \alpha_2 p = g(p)$. Take $p \notin [q_1, q_2]$: the limit is $0 = g(p)$. At $p = q^*$ each of the two pieces tends to one half of its bracket. The brackets are $\alpha_1 q^* - \beta_1$ and $\beta_2 - \alpha_2 q^*$, and both equal $g(q^*)$ by continuity of $V - \chi$. So the limit is $g(q^*)$. At $p = q_1 > 0$ the limit is $\frac{1}{2}(\alpha_1 q_1 - \beta_1) = 0 = g(q_1)$. At $p = q_2 < 1$ it is $\frac{1}{2}(\beta_2 - \alpha_2 q_2) = 0 = g(q_2)$. These are the confounded values $\mathbb{K}[p \in S]$ and $(V - \chi)^+(p)$ of Corollary 1(ii)–(iii). $\square$

Appendix B. Results used from Paper 1

The three statements below are restated from Paper 1 [1], a working paper, with the symbols of this paper; their proofs are in Paper 1. Throughout this paper, “Paper 1, Proposition 1” refers to the last of them and “Proposition 1” to this paper’s Gaussian result of Section 4.8.

Definition D8 of [1] (Loss table). The general two-action, two-cause loss table $L(a, M)$, $a \in \{\text{contain}, \text{leave}\}$, $M \in \{K, F\}$, is

$$\begin{aligned} L(\text{contain}, K) &= C_K, & L(\text{contain}, F) &= C_F, \\ L(\text{leave}, K) &= H_K, & L(\text{leave}, F) &= H_F, \end{aligned}$$

with $C_K, C_F, H_K, H_F \in \mathbb{R}$. The table of [1] (its Section 3) is the special case $H_F = 0$, in which its routine containment cost, disruption cost and harm are C_K , C_F and H_K . Its Proposition 1 is stated for that special case.

Definition D11 of [1] (Gaussian discriminability). If $Y \mid M \sim \mathcal{N}(\mu_M, \Sigma_Y)$, $M \in \{K, F\}$, with a common covariance $\Sigma_Y \succ 0$, let $\delta = \mu_K - \mu_F$ and $d = \sqrt{\delta^\top \Sigma_Y^{-1} \delta} \geq 0$ (the whitened mean separation).

Proposition 1 of [1] (Cost of ambiguity is bounded below under confounding). Suppose condition (*) of Theorem 1 of [1] holds (the confounded case, in which the laws of the observation Y under K and under F

coincide), $0 < p < 1$, $C_F > 0$, and $H_K > C_K$ (harm strictly worse than routine containment), with $H_F = 0$ (Definition D8). Then the best achievable expected cost among policies mapping the observations to $\{\text{contain, leave}\}$ is

$$R^* = \min\{p C_K + (1 - p) C_F, \quad p H_K\},$$

the oracle's expected cost is $R_{\text{oracle}} = p C_K$, and

$$\text{CoA}^* = R^* - R_{\text{oracle}} = \min\{(1 - p) C_F, \quad p (H_K - C_K)\} > 0,$$

that is, $\text{CoA}^* = \min\{p a, (1 - p) b\}$ with $a = H_K - C_K$ and $b = C_F - H_F = C_F$.

References

- [1] C. Mejrissi, Attack-fault identifiability and the cost of ambiguity at the electronic trading gateway, working paper (2026).
- [2] S. Tariq, M. Baruwat Chhetri, S. Nepal, C. Paris, Alert fatigue in security operations centres: Research challenges and opportunities, *ACM Computing Surveys* 57 (9) (2025) 224:1–224:38. doi:10.1145/3723158.
- [3] J. Ghadermazi, A. Shah, S. Jajodia, A machine learning and optimization framework for efficient alert management in a cybersecurity operations center, *Digital Threats: Research and Practice* 5 (2) (2024) 1–23. doi:10.1145/3644393.
- [4] S. Wiefeling, P. R. Jørgensen, S. Thunem, L. Lo Iacono, Pump up password security! Evaluating and enhancing risk-based authentication on a real-world large-scale online service, *ACM Transactions on Privacy and Security* 26 (1) (2023) 6:1–6:36. doi:10.1145/3546069.
- [5] S. G. Pauker, J. P. Kassirer, The threshold approach to clinical decision making, *New England Journal of Medicine* 302 (20) (1980) 1109–1117. doi:10.1056/NEJM198005153022003.
- [6] H. Raiffa, R. Schlaifer, *Applied Statistical Decision Theory*, Studies in Managerial Economics, Division of Research, Graduate School of Business Administration, Harvard University, Boston, 1961.
- [7] M. C. Mungan, The undertesting bias in clinical decision making: The role of test dichotomization, *Medical Decision Making* 46 (7) (2026) 819–824. doi:10.1177/0272989X261471178.

- [8] E. Horvitz, G. Rutledge, Time-dependent utility and action under uncertainty, in: *Proceedings of the Seventh Conference on Uncertainty in Artificial Intelligence (UAI)*, 1991, pp. 151–158.
- [9] E. Horvitz, A. Seiver, Time-critical action: Representations and application, in: *Proceedings of the Thirteenth Conference on Uncertainty in Artificial Intelligence (UAI)*, 1997, pp. 250–257.
- [10] L. F. Schroeder, P. Rebman, P. Kasaie, E. Kenu, J. Zelner, D. W. Dowdy, A generalizable decision-making framework for selecting onsite versus send-out clinical laboratory testing, *Medical Decision Making* 44 (3) (2024) 307–319. doi:10.1177/0272989X241232666.
- [11] S. Eckermann, A. R. Willan, Time and expected value of sample information wait for no patient, *Value in Health* 11 (3) (2008) 522–526.
- [12] M. Kremer, F. de Véricourt, Mismanaging diagnostic accuracy under congestion, *Operations Research* 71 (3) (2023) 895–916. doi:10.1287/opre.2022.2292.
- [13] D. Golovin, A. Krause, D. Ray, Near-optimal Bayesian active learning with noisy observations, in: *Advances in Neural Information Processing Systems* 23, 2010, pp. 766–774.
- [14] R. Xu, T. Zariphopoulou, L. Zhang, Decision making under costly sequential information acquisition: The paradigm of reversible and irreversible decisions, *arXiv preprint* (2024). arXiv:2401.00569v4.
- [15] E. Lehrer, T. Wang, The value of information in stopping problems, *Economic Theory* 78 (2) (2024) 619–648. doi:10.1007/s00199-023-01543-8.
- [16] W. Lee, W. Fan, M. Miller, S. J. Stolfo, E. Zadok, Toward cost-sensitive modeling for intrusion detection and response, *Journal of Computer Security* 10 (1–2) (2002) 5–22. doi:10.3233/JCS-2002-101-202.
- [17] K. Hammar, R. Stadler, Intrusion prevention through optimal stopping, *IEEE Transactions on Network and Service Management* 19 (3) (2022) 2333–2348. doi:10.1109/TNSM.2022.3176781.

- [18] K. Hammar, R. Stadler, Learning near-optimal intrusion responses against dynamic attackers, *IEEE Transactions on Network and Service Management* 21 (1) (2024) 1158–1177. doi:10.1109/TNSM.2023.3293413.
- [19] K. Hammar, T. Li, R. Stadler, Q. Zhu, Adaptive security response strategies through conjectural online learning, *IEEE Transactions on Information Forensics and Security* 20 (2025) 4055–4070. doi:10.1109/TIFS.2025.3558600.
- [20] C. Kruegel, W. Robertson, Alert verification: Determining the success of intrusion attempts, in: *Detection of Intrusions and Malware & Vulnerability Assessment (DIMVA 2004)*, Lecture Notes in Informatics, Gesellschaft für Informatik, 2004.
- [21] Y. Kim, G. Dán, Q. Zhu, Human-in-the-loop cyber intrusion detection using active learning, *IEEE Transactions on Information Forensics and Security* 19 (2024) 8658–8672. doi:10.1109/TIFS.2024.3434647.
- [22] Y. Kim, G. Dán, Dynamic alert prioritization for real-time situational awareness: A hidden Markov model framework with active learning, *IEEE Transactions on Dependable and Secure Computing* 22 (6) (2025) 6913–6928. doi:10.1109/TDSC.2025.3592127.
- [23] F. Jalalvand, M. Baruwat Chhetri, S. Nepal, C. Paris, Adaptive alert prioritisation in security operations centres via learning to defer with human feedback, *Information Sciences* 760 (2027) 124128. doi:10.1016/j.ins.2026.124128.
- [24] F. Şahin, N. S. Mert, Dual-threshold conformal deferral for trustworthy security alert triage, *Electronics* 15 (18) (2026) 4084. doi:10.3390/electronics15184084.
- [25] K. Zhang, C. Keliris, T. Parisini, M. M. Polycarpou, Identification of sensor replay attacks and physical faults for cyber-physical systems, *IEEE Control Systems Letters* 6 (2022) 1178–1183. doi:10.1109/LCSYS.2021.3089674.
- [26] M. S. Ahsan, H. Wang, V. R. Motakatla, M. Zhu, P. Liu, Differentiation between faults and cyberattacks through combined analysis of cy-

- berspace logs and physical measurements, *IEEE Transactions on Smart Grid* (2026). doi:10.1109/TSG.2026.3728276.
- [27] M. Saelens, P. Latinne, C. Decaestecker, Adjusting the outputs of a classifier to new a priori probabilities: A simple procedure, *Neural Computation* 14 (1) (2002) 21–41. doi:10.1162/089976602753284446.
 - [28] C. Elkan, The foundations of cost-sensitive learning, in: *Proceedings of the 17th International Joint Conference on Artificial Intelligence (IJCAI)*, 2001, pp. 973–978.
 - [29] S. Höppner, B. Baesens, W. Verbeke, T. Verdonck, Instance-dependent cost-sensitive learning for detecting transfer fraud, *European Journal of Operational Research* 297 (1) (2022) 291–300. doi:10.1016/j.ejor.2021.05.028.
 - [30] T. Silva Filho, H. Song, M. Perello-Nieto, R. Santos-Rodriguez, M. Kull, P. Flach, Classifier calibration: A survey on how to assess and improve predicted class probabilities, *Machine Learning* 112 (9) (2023) 3211–3260. doi:10.1007/s10994-023-06336-7.
 - [31] D. Byrd, M. Hybinette, T. H. Balch, ABIDES: Towards high-fidelity multi-agent market simulation, in: *Proceedings of the 2020 ACM SIGSIM Conference on Principles of Advanced Discrete Simulation (PADS)*, 2020, pp. 11–22. doi:10.1145/3384441.3395986.

Table 3: Primary exploratory test per failure cell, with Paper 1’s cost matrix and, below it, with the measured cost of leaving a fault (sensitivity, decides nothing). Counts of plausible grid points predicted in, out and in the band; agreeing, disagreeing and unresolved points; the agreement share against $\bar{\sigma} = 0.9$. Last column: whether targeted verification beats the baseline at some implausible point. Designs as in Paper 1 (V1: V1_original, Study4: Study4_expanded21ch, V2: V2_study5, V3: V3_third, V4: V4_fourth); C: contrast, SM: summary-matched regime.

q^*	Design	Regime	p	Baseline	In	Out	Band	Agree	Disagree	Unres.	Share	Agrees	Unres. cell	Impl. $U < 0$
Paper 1	V1	C	0.05	leave	120	67	341	181	2	4	0.989	yes	no	yes
	V2	C	0.05	leave	103	214	211	299	1	17	0.997	yes	no	yes
	V2	SM	0.05	leave	52	417	59	448	5	16	0.989	yes	no	yes
	V2	SM	0.10	leave	162	242	124	365	2	37	0.995	yes	no	yes
	V2	SM	0.20	leave	377	69	82	399	2	45	0.995	yes	no	yes
	V2	SM	0.30	leave	461	67	0	492	0	36	1.000	yes	no	yes
	pooled (decides nothing)							2184	12		0.995			
measured H_F	V1	C	0.05	leave	129	67	332	188	1	7	0.995	yes	no	yes
	V2	C	0.05	leave	103	213	212	297	1	18	0.997	yes	no	yes
	V2	SM	0.05	leave	52	416	60	447	5	16	0.989	yes	no	yes
	V2	SM	0.10	leave	162	242	124	365	3	36	0.992	yes	no	yes
	V2	SM	0.20	leave	377	69	82	397	2	47	0.995	yes	no	yes
	V2	SM	0.30	leave	461	67	0	492	0	36	1.000	yes	no	yes
	pooled (decides nothing)							2186	12		0.995			

Table 4: Policy comparison per cell: number of plausible grid points at which targeted verification is significantly worse / significantly better than each other policy (95% bootstrap intervals, 2000 resamples), with the median false-positive and false-negative rates of targeted verification and of verify-everything over the plausible points. Designs as in Paper 1 (V1: V1_original, Study4: Study4_expanded21ch, V2: V2_study5, V3: V3_third, V4: V4_fourth); C: contrast, SM: summary-matched regime.

Design	Regime	p	Failure	vs always contain	vs always leave	vs passive	vs verify all	Targeted FPR, FNR	Verify-all FPR, FNR
V1	C	0.05	yes	0 / 528	0 / 150	0 / 201	0 / 528	0.010, 0.873	0.200, 0.150
V1	C	0.10	no	0 / 528	0 / 286	0 / 306	0 / 508	0.025, 0.637	0.200, 0.150
V1	C	0.20	no	0 / 528	0 / 528	0 / 334	0 / 439	0.087, 0.364	0.200, 0.150
V1	C	0.30	no	0 / 528	0 / 528	0 / 398	0 / 423	0.161, 0.226	0.200, 0.150
V1	C	0.50	no	0 / 528	0 / 528	0 / 307	0 / 439	0.342, 0.088	0.200, 0.150
V1	SM	0.05	no	0 / 528	0 / 528	0 / 84	0 / 528	0.008, 0.345	0.200, 0.150
V1	SM	0.10	no	0 / 528	0 / 528	5 / 162	0 / 528	0.023, 0.246	0.200, 0.150
V1	SM	0.20	no	0 / 528	0 / 528	0 / 318	0 / 527	0.049, 0.177	0.200, 0.150
V1	SM	0.30	no	0 / 528	0 / 528	0 / 382	0 / 526	0.083, 0.137	0.200, 0.150
V1	SM	0.50	no	0 / 528	0 / 528	0 / 277	0 / 520	0.142, 0.088	0.200, 0.150
Study4	C	0.05	no	0 / 528	0 / 528	0 / 71	0 / 528	0.000, 0.024	0.200, 0.150
Study4	C	0.10	no	0 / 528	0 / 528	1 / 0	0 / 528	0.000, 0.013	0.200, 0.150
Study4	C	0.20	no	0 / 528	0 / 528	315 / 0	0 / 528	0.000, 0.005	0.200, 0.150
Study4	C	0.30	no	0 / 528	0 / 528	175 / 0	0 / 528	0.000, 0.003	0.200, 0.150
Study4	C	0.50	no	0 / 528	0 / 528	0 / 0	0 / 528	0.001, 0.003	0.200, 0.150
Study4	SM	0.05	no	0 / 528	0 / 528	4 / 33	0 / 528	0.003, 0.248	0.200, 0.150
Study4	SM	0.10	no	0 / 528	0 / 528	0 / 228	0 / 528	0.013, 0.175	0.200, 0.150
Study4	SM	0.20	no	0 / 528	0 / 528	0 / 266	0 / 528	0.035, 0.118	0.200, 0.150
Study4	SM	0.30	no	0 / 528	0 / 528	0 / 277	0 / 528	0.057, 0.088	0.200, 0.150
Study4	SM	0.50	no	0 / 528	0 / 528	0 / 282	0 / 528	0.108, 0.056	0.200, 0.150
V2	C	0.05	yes	0 / 528	0 / 109	0 / 383	0 / 516	0.004, 0.904	0.200, 0.150
V2	C	0.10	no	0 / 528	0 / 371	0 / 245	0 / 496	0.013, 0.716	0.200, 0.150
V2	C	0.20	no	0 / 528	0 / 528	0 / 369	0 / 426	0.064, 0.401	0.200, 0.150
V2	C	0.30	no	0 / 528	0 / 528	0 / 350	0 / 399	0.158, 0.249	0.200, 0.150
V2	C	0.50	no	0 / 528	0 / 528	0 / 331	0 / 464	0.338, 0.098	0.200, 0.150
V2	SM	0.05	yes	0 / 528	0 / 31	0 / 31	0 / 506	0.000, 1.000	0.200, 0.150
V2	SM	0.10	yes	0 / 528	0 / 123	0 / 123	0 / 435	0.003, 0.987	0.200, 0.150
V2	SM	0.20	yes	0 / 528	0 / 330	0 / 341	0 / 259	0.051, 0.415	0.200, 0.150
V2	SM	0.30	yes	0 / 528	0 / 425	0 / 443	0 / 103	0.100, 0.200	0.200, 0.150
V2	SM	0.50	no	0 / 367	0 / 528	0 / 359	0 / 217	0.307, 0.047	0.200, 0.150
V3	C	0.05	no	0 / 528	0 / 528	0 / 21	0 / 528	0.000, 0.027	0.200, 0.150
V3	C	0.10	no	0 / 528	0 / 528	8 / 0	0 / 528	0.000, 0.014	0.200, 0.150
V3	C	0.20	no	0 / 528	0 / 528	0 / 259	0 / 528	0.001, 0.004	0.200, 0.150
V3	C	0.30	no	0 / 528	0 / 528	15 / 0	0 / 528	0.001, 0.001	0.200, 0.150
V3	C	0.50	no	0 / 528	0 / 528	64 / 0	0 / 528	0.003, 0.000	0.200, 0.150
V3	SM	0.05	no	0 / 528	0 / 528	0 / 95	0 / 528	0.005, 0.337	0.200, 0.150
V3	SM	0.10	no	0 / 528	0 / 528	0 / 225	0 / 528	0.014, 0.251	0.200, 0.150
V3	SM	0.20	no	0 / 528	0 / 528	0 / 288	0 / 528	0.034, 0.173	0.200, 0.150
V3	SM	0.30	no	0 / 528	0 / 528	0 / 255	0 / 528	0.054, 0.131	0.200, 0.150
V3	SM	0.50	no	0 / 528	0 / 528	0 / 337	0 / 526	0.102, 0.077	0.200, 0.150
V4	C	0.05	no	0 / 528	0 / 314	0 / 211	0 / 528	0.007, 0.719	0.200, 0.150
V4	C	0.10	no	0 / 528	0 / 443	0 / 293	0 / 522	0.030, 0.515	0.200, 0.150
V4	C	0.20	no	0 / 528	0 / 528	0 / 347	0 / 499	0.092, 0.290	0.200, 0.150
V4	C	0.30	no	0 / 528	0 / 528	0 / 459	0 / 498	0.157, 0.192	0.200, 0.150
V4	C	0.50	no	0 / 528	0 / 528	0 / 383	0 / 498	0.323, 0.083	0.200, 0.150
V4	SM	0.05	no	0 / 528	0 / 528	215 / 0	0 / 528	0.000, 0.001	0.200, 0.150
V4	SM	0.10	no	0 / 528	0 / 528	89 / 0	0 / 528	0.000, 0.000	0.200, 0.150
V4	SM	0.20	no	0 / 528	0 / 528	31 / 0	0 / 528	0.000, 0.000	0.200, 0.150
V4	SM	0.30	no	0 / 528	0 / 528	28 / 0	0 / 528	0.000, 0.000	0.200, 0.150
V4	SM	0.50	no	0 / 528	0 / 528	68 / 0	0 / 528	0.000, 0.000	0.200, 0.150

Table 5: Cells: Paper 1’s failure cells and best constant policy; cost of the passive rule (with its false-positive and false-negative rates), of the best constant policy and the passive cost of ambiguity with its interval; the corrected-posterior quantiles at α_{in} , the band level and α_{out} ; the median corrected logit posterior minus logit q^* . Costs in Paper 1’s monetary units. Designs as in Paper 1 (V1: V1_original, Study4: Study4_expanded21ch, V2: V2_study5, V3: V3_third, V4: V4_fourth); C: contrast, SM: summary-matched regime.

Design	Regime	p	Failure	Best const.	Passive cost (FPR, FNR)	Best-const. cost	CoA* [CI]	$Q_{\alpha_{in}}$	Q_{band}	$Q_{\alpha_{out}}$	Δlogit
V1	C	0.05	yes	leave	68,724 (0.010, 0.960)	65,875	44,178 [39,809, 50,047]	0.106	0.172	0.427	-2.79
V1	C	0.10	no	leave	131,189 (0.020, 0.890)	131,750	82,095 [75,533, 89,711]	0.218	0.313	0.612	-2.01
V1	C	0.20	no	leave	239,604 (0.087, 0.657)	263,500	141,416 [126,925, 155,842]	0.427	0.531	0.789	-1.10
V1	C	0.30	no	leave	327,165 (0.213, 0.440)	395,250	179,884 [159,716, 201,108]	0.590	0.672	0.867	-0.46
V1	C	0.50	no	contain	422,810 (0.510, 0.137)	482,448	177,341 [156,249, 198,208]	0.798	0.861	0.950	0.56
V1	SM	0.05	no	leave	46,683 (0.010, 0.427)	65,875	22,136 [17,220, 28,033]	0.139	0.270	0.845	-5.15
V1	SM	0.10	no	leave	87,150 (0.023, 0.340)	131,750	38,056 [29,976, 47,183]	0.332	0.649	0.967	-4.25
V1	SM	0.20	no	leave	163,788 (0.080, 0.213)	263,500	65,600 [52,600, 79,864]	0.806	0.950	0.992	-3.06
V1	SM	0.30	no	leave	229,516 (0.113, 0.180)	395,250	82,235 [66,228, 97,963]	0.950	0.984	0.996	-2.03
V1	SM	0.50	no	contain	335,349 (0.170, 0.120)	482,448	89,880 [71,745, 108,415]	0.990	0.996	0.999	0.30
Study4	C	0.05	no	leave	26,200 (0.000, 0.040)	65,875	1,653 [827, 2,617]	0.002	0.030	0.998	-8.81
Study4	C	0.10	no	leave	50,471 (0.000, 0.017)	131,750	1,378 [276, 2,755]	0.061	0.989	1.000	-7.86
Study4	C	0.20	no	leave	98,739 (0.000, 0.003)	263,500	551 [0, 1,653]	0.995	0.999	1.000	-6.54
Study4	C	0.30	no	leave	148,108 (0.000, 0.003)	395,250	827 [0, 2,480]	0.999	1.000	1.000	-5.54
Study4	C	0.50	no	contain	247,636 (0.003, 0.003)	482,448	2,168 [0, 5,713]	1.000	1.000	1.000	-0.67
Study4	SM	0.05	no	leave	37,496 (0.000, 0.313)	65,875	12,949 [10,745, 15,154]	0.081	0.227	0.953	-5.88
Study4	SM	0.10	no	leave	76,316 (0.017, 0.243)	131,750	27,222 [20,432, 34,795]	0.309	0.854	0.990	-4.86
Study4	SM	0.20	no	leave	143,596 (0.050, 0.160)	263,500	45,408 [33,806, 56,849]	0.929	0.986	0.999	-3.57
Study4	SM	0.30	no	leave	203,020 (0.073, 0.127)	395,250	55,739 [42,223, 69,546]	0.983	0.995	0.999	-2.39
Study4	SM	0.50	no	contain	310,533 (0.147, 0.073)	482,448	65,064 [49,266, 80,848]	0.997	0.999	1.000	0.52
V2	C	0.05	yes	leave	66,963 (0.003, 0.990)	65,875	42,416 [40,502, 45,693]	0.096	0.125	0.206	-2.74
V2	C	0.10	no	leave	127,937 (0.003, 0.937)	131,750	78,843 [75,493, 82,789]	0.188	0.241	0.355	-1.96
V2	C	0.20	no	leave	242,328 (0.063, 0.727)	263,500	144,141 [130,884, 157,624]	0.376	0.452	0.629	-1.08
V2	C	0.30	no	leave	331,007 (0.200, 0.473)	395,250	183,726 [162,748, 203,895]	0.528	0.620	0.782	-0.48
V2	C	0.50	no	contain	439,196 (0.573, 0.140)	482,448	193,727 [171,831, 214,636]	0.765	0.820	0.895	0.50
V2	SM	0.05	yes	leave	65,875 (0.000, 1.000)	65,875	41,328 [41,328, 41,328]	0.066	0.075	0.100	-2.44
V2	SM	0.10	yes	leave	131,750 (0.000, 1.000)	131,750	82,656 [82,656, 82,656]	0.129	0.145	0.191	-1.69
V2	SM	0.20	yes	leave	266,190 (0.010, 0.993)	263,500	168,002 [164,201, 173,058]	0.250	0.278	0.346	-0.88
V2	SM	0.30	yes	leave	397,833 (0.090, 0.890)	395,250	250,552 [237,286, 264,376]	0.364	0.398	0.476	-0.33
V2	SM	0.50	no	contain	481,253 (0.983, 0.007)	482,448	235,785 [231,247, 241,112]	0.572	0.613	0.679	0.52
V3	C	0.05	no	leave	26,200 (0.000, 0.040)	65,875	1,653 [827, 2,617]	0.002	0.110	1.000	-9.10
V3	C	0.10	no	leave	50,471 (0.000, 0.017)	131,750	1,378 [276, 2,480]	0.208	0.995	1.000	-8.11
V3	C	0.20	no	leave	101,656 (0.003, 0.013)	263,500	3,468 [551, 7,098]	0.998	1.000	1.000	-6.83
V3	C	0.30	no	leave	148,387 (0.003, 0.000)	395,250	1,106 [0, 3,318]	1.000	1.000	1.000	-5.86
V3	C	0.50	no	contain	246,259 (0.003, 0.000)	482,448	790 [0, 2,370]	1.000	1.000	1.000	0.89
V3	SM	0.05	no	leave	42,992 (0.003, 0.410)	65,875	18,445 [15,291, 22,412]	0.074	0.211	0.946	-5.62
V3	SM	0.10	no	leave	82,653 (0.017, 0.320)	131,750	33,559 [26,583, 41,452]	0.271	0.774	0.991	-4.74
V3	SM	0.20	no	leave	154,293 (0.043, 0.240)	263,500	56,106 [45,053, 67,839]	0.851	0.978	0.998	-3.54
V3	SM	0.30	no	leave	214,860 (0.067, 0.183)	395,250	67,579 [53,772, 82,748]	0.974	0.995	0.999	-2.42
V3	SM	0.50	no	contain	325,282 (0.133, 0.117)	482,448	79,813 [63,263, 98,334]	0.997	0.999	1.000	0.18
V4	C	0.05	no	leave	63,780 (0.007, 0.877)	65,875	39,233 [35,680, 44,562]	0.121	0.199	0.370	-2.91
V4	C	0.10	no	leave	122,781 (0.030, 0.737)	131,750	73,687 [64,649, 83,465]	0.245	0.376	0.655	-2.07
V4	C	0.20	no	leave	222,714 (0.113, 0.493)	263,500	124,526 [108,809, 141,090]	0.513	0.673	0.852	-1.16
V4	C	0.30	no	leave	309,551 (0.230, 0.347)	395,250	162,270 [142,118, 183,259]	0.688	0.804	0.916	-0.48
V4	C	0.50	no	contain	411,770 (0.487, 0.123)	482,448	166,301 [145,576, 187,573]	0.887	0.933	0.973	0.66
V4	SM	0.05	no	leave	24,547 (0.000, 0.000)	65,875	0 [0, 0]	0.001	0.007	0.989	-10.58
V4	SM	0.10	no	leave	49,094 (0.000, 0.000)	131,750	0 [0, 0]	0.014	0.983	0.998	-9.64
V4	SM	0.20	no	leave	98,188 (0.000, 0.000)	263,500	0 [0, 0]	0.992	0.997	1.000	-8.28
V4	SM	0.30	no	leave	147,281 (0.000, 0.000)	395,250	0 [0, 0]	0.997	0.999	1.000	-7.07
V4	SM	0.50	no	contain	245,469 (0.000, 0.000)	482,448	0 [0, 0]	0.999	1.000	1.000	-1.51

Table 6: Verification mass, saving and recovered share per cell over the plausible grid points: number of points with a non-empty interval; m_c ; number of points at which the realised saving $\hat{D}_c$ is significantly positive / significantly negative / not distinguishable from zero; medians of $\hat{D}_c$ and of the plug-in saving $\tilde{D}_c$; number of points with $\tilde{D}_c$ significantly positive; ρ_c (n.r.: not reported, cost of ambiguity indistinguishable from zero). Descriptive; decides no claim. Designs as in Paper 1 (V1: V1_original, Study4: Study4_expanded21ch, V2: V2_study5, V3: V3_third, V4: V4_fourth); C: contrast, SM: summary-matched regime.

Design	Regime	p	Failure	Non-empty	m_c	median [min, max]	$\hat{D}_c$: > 0 / < 0 / covers 0	$\hat{D}_c$ median	$\tilde{D}_c$ median	$\tilde{D}_c$ > 0	ρ_c	median [min, max]
V1	C	0.05	yes	461	0.042	[0.000, 0.638]	201 / 0 / 327	2,875	1,732	453	0.065	[-0.019, 0.629]
V1	C	0.10	no	461	0.143	[0.000, 0.878]	306 / 0 / 222	7,521	6,934	461	0.092	[-0.044, 0.767]
V1	C	0.20	no	461	0.411	[0.000, 0.988]	334 / 0 / 194	21,918	23,246	461	0.155	[-0.016, 0.857]
V1	C	0.30	no	461	0.572	[0.000, 0.992]	398 / 0 / 130	37,200	34,915	461	0.207	[-0.011, 0.885]
V1	C	0.50	no	461	0.522	[0.000, 0.993]	307 / 0 / 221	26,033	31,820	461	0.147	[-0.043, 0.875]
V1	SM	0.05	no	461	0.052	[0.000, 0.222]	84 / 0 / 444	1,545	2,877	461	0.070	[-0.037, 0.590]
V1	SM	0.10	no	461	0.100	[0.000, 0.331]	162 / 5 / 361	2,683	6,118	461	0.070	[-0.070, 0.661]
V1	SM	0.20	no	461	0.133	[0.000, 0.459]	318 / 0 / 210	9,723	7,295	461	0.148	[-0.025, 0.745]
V1	SM	0.30	no	461	0.136	[0.000, 0.537]	382 / 0 / 146	12,348	7,282	461	0.150	[0.000, 0.772]
V1	SM	0.50	no	461	0.147	[0.000, 0.575]	277 / 0 / 251	9,975	8,381	461	0.111	[-0.009, 0.778]
Study4	C	0.05	no	461	0.003	[0.000, 0.026]	71 / 0 / 457	370	171	454	0.224	[-0.114, 0.788]
Study4	C	0.10	no	461	0.004	[0.000, 0.044]	0 / 1 / 527	0	234	409	0.000	[-0.414, 0.574]
Study4	C	0.20	no	461	0.003	[0.000, 0.061]	0 / 315 / 213	-743	211	333	n.r.	
Study4	C	0.30	no	461	0.004	[0.000, 0.064]	0 / 175 / 353	-502	97	227	n.r.	
Study4	C	0.50	no	461	0.003	[0.000, 0.080]	0 / 0 / 528	503	256	206	n.r.	
Study4	SM	0.05	no	461	0.035	[0.000, 0.188]	33 / 4 / 491	-681	1,895	460	-0.053	[-0.178, 0.500]
Study4	SM	0.10	no	461	0.066	[0.000, 0.270]	228 / 0 / 300	3,760	4,239	461	0.138	[-0.020, 0.663]
Study4	SM	0.20	no	461	0.108	[0.000, 0.368]	266 / 0 / 262	6,784	5,888	461	0.149	[0.000, 0.746]
Study4	SM	0.30	no	461	0.124	[0.000, 0.416]	277 / 0 / 251	7,539	7,171	461	0.135	[-0.008, 0.754]
Study4	SM	0.50	no	461	0.121	[0.000, 0.438]	282 / 0 / 246	8,918	6,932	461	0.137	[-0.004, 0.793]
V2	C	0.05	yes	461	0.015	[0.000, 0.691]	383 / 0 / 145	1,725	332	436	0.041	[-0.001, 0.570]
V2	C	0.10	no	461	0.113	[0.000, 0.927]	245 / 0 / 283	4,124	4,986	461	0.052	[-0.014, 0.749]
V2	C	0.20	no	461	0.425	[0.000, 0.992]	369 / 0 / 159	22,825	23,695	461	0.158	[-0.011, 0.850]
V2	C	0.30	no	461	0.641	[0.000, 0.998]	350 / 0 / 178	37,162	39,694	461	0.202	[-0.004, 0.881]
V2	C	0.50	no	461	0.578	[0.000, 0.998]	331 / 0 / 197	33,588	36,927	461	0.173	[-0.036, 0.885]
V2	SM	0.05	yes	461	0.000	[0.000, 1.000]	31 / 0 / 497	0	0	141	0.000	[-0.026, 0.517]
V2	SM	0.10	yes	461	0.010	[0.000, 1.000]	123 / 0 / 405	0	303	336	0.000	[-0.025, 0.756]
V2	SM	0.20	yes	461	0.619	[0.000, 1.000]	341 / 0 / 187	20,831	17,007	453	0.124	[-0.004, 0.878]
V2	SM	0.30	yes	461	0.990	[0.000, 1.000]	443 / 0 / 85	82,796	70,951	461	0.330	[-0.012, 0.917]
V2	SM	0.50	no	461	0.879	[0.000, 1.000]	359 / 0 / 169	35,558	47,744	461	0.151	[0.000, 0.909]
V3	C	0.05	no	461	0.003	[0.000, 0.023]	21 / 0 / 507	178	177	438	0.107	[-0.038, 0.806]
V3	C	0.10	no	461	0.007	[0.000, 0.035]	0 / 8 / 520	-177	270	410	-0.128	[-0.681, 0.699]
V3	C	0.20	no	461	0.006	[0.000, 0.055]	259 / 0 / 269	2,049	600	445	0.591	[0.000, 0.825]
V3	C	0.30	no	461	0.006	[0.000, 0.065]	0 / 15 / 513	-257	372	345	n.r.	
V3	C	0.50	no	461	0.003	[0.000, 0.080]	0 / 64 / 464	-258	51	186	n.r.	
V3	SM	0.05	no	461	0.030	[0.000, 0.188]	95 / 0 / 433	0	1,428	461	0.000	[-0.072, 0.568]
V3	SM	0.10	no	461	0.062	[0.000, 0.278]	225 / 0 / 303	3,849	4,096	461	0.115	[-0.033, 0.681]
V3	SM	0.20	no	461	0.114	[0.000, 0.385]	288 / 0 / 240	8,430	6,174	461	0.150	[-0.022, 0.726]
V3	SM	0.30	no	461	0.134	[0.000, 0.448]	255 / 0 / 273	7,282	8,154	461	0.108	[-0.009, 0.770]
V3	SM	0.50	no	461	0.148	[0.000, 0.520]	337 / 0 / 191	14,026	8,236	461	0.176	[-0.013, 0.799]
V4	C	0.05	no	461	0.058	[0.000, 0.552]	211 / 0 / 317	2,119	3,400	461	0.054	[-0.052, 0.619]
V4	C	0.10	no	461	0.152	[0.000, 0.787]	293 / 0 / 235	8,750	8,187	461	0.119	[-0.026, 0.750]
V4	C	0.20	no	461	0.370	[0.000, 0.925]	347 / 0 / 181	20,055	21,561	461	0.161	[0.000, 0.843]
V4	C	0.30	no	461	0.478	[0.000, 0.968]	459 / 0 / 69	36,047	29,120	461	0.222	[0.000, 0.875]
V4	C	0.50	no	461	0.429	[0.000, 0.973]	383 / 0 / 145	28,192	24,189	461	0.170	[-0.011, 0.875]
V4	SM	0.05	no	461	0.000	[0.000, 0.035]	0 / 215 / 313	-117	33	215	n.r.	
V4	SM	0.10	no	461	0.000	[0.000, 0.045]	0 / 89 / 439	-68	2	89	n.r.	
V4	SM	0.20	no	461	0.000	[0.000, 0.039]	0 / 31 / 497	0	0	31	n.r.	
V4	SM	0.30	no	461	0.000	[0.000, 0.031]	0 / 28 / 500	0	0	28	n.r.	
V4	SM	0.50	no	461	0.000	[0.000, 0.028]	0 / 68 / 460	0	0	68	n.r.	

Table 7: Critical delay $r_K\tau^*/a$ (Theorem 2) per sensitivity, specificity and cost of a check, in units of a/r_K (–: the interval is empty at every delay), and the number of (x, y) points with a non-empty interval. Closed forms agree with the numerical search at every point.

s	t	$x = 0.001$	$x = 0.01$	$x = 0.05$	$x = 0.1$	$x = 0.2$	$x = 0.5$	Non-empty of 36
0.7	0.7	0.399	0.390	0.350	0.300	0.200	–	24
0.7	0.8	0.499	0.490	0.450	0.400	0.300	–	25
0.7	0.9	0.599	0.590	0.550	0.500	0.400	0.100	31
0.7	0.99	0.689	0.680	0.640	0.590	0.490	0.190	33
0.8	0.7	0.499	0.490	0.450	0.400	0.300	–	25
0.8	0.8	0.599	0.590	0.550	0.500	0.400	0.100	31
0.8	0.9	0.699	0.690	0.650	0.600	0.500	0.200	33
0.8	0.99	0.789	0.780	0.740	0.690	0.590	0.290	35
0.9	0.7	0.599	0.590	0.550	0.500	0.400	0.100	31
0.9	0.8	0.699	0.690	0.650	0.600	0.500	0.200	33
0.9	0.9	0.799	0.790	0.750	0.700	0.600	0.300	35
0.9	0.99	0.889	0.880	0.840	0.790	0.690	0.390	35
0.99	0.7	0.689	0.680	0.640	0.590	0.490	0.190	33
0.99	0.8	0.789	0.780	0.740	0.690	0.590	0.290	35
0.99	0.9	0.889	0.880	0.840	0.790	0.690	0.390	35
0.99	0.99	0.979	0.970	0.930	0.880	0.780	0.480	35

Table 8: Sensitivity analysis: passive threshold with the measured cost of leaving a fault, and, over the plausible grid points, the median verification mass, the number of points with a significantly positive realised saving, and the number at which targeted verification is significantly cheaper than the best constant policy, under each threshold. Decides nothing. Designs as in Paper 1 (V1: V1_original, Study4: Study4_expanded21ch, V2: V2_study5, V3: V3_third, V4: V4_fourth); C: contrast, SM: summary-matched regime.

Design	Regime	p	Failure	q^*	m_c median		$\widehat{D}_c > 0$		Targeted < best constant	
					measured	primary	measured	primary	measured	primary
V1	C	0.05	yes		0.348	0.042	0.049	201	199	150
V1	C	0.10	no		0.348	0.143	0.160	306	305	286
V1	C	0.20	no		0.348	0.411	0.425	334	338	528
V1	C	0.30	no		0.348	0.572	0.575	398	366	528
V1	C	0.50	no		0.348	0.522	0.508	307	287	528
V1	SM	0.05	no		0.309	0.052	0.063	84	114	528
V1	SM	0.10	no		0.309	0.100	0.100	162	263	528
V1	SM	0.20	no		0.309	0.133	0.126	318	322	528
V1	SM	0.30	no		0.309	0.136	0.130	382	299	528
V1	SM	0.50	no		0.309	0.147	0.142	277	267	528
Study4	C	0.05	no		0.348	0.003	0.002	71	116	528
Study4	C	0.10	no		0.348	0.004	0.004	0	2	528
Study4	C	0.20	no		0.348	0.003	0.003	0	0	528
Study4	C	0.30	no		0.348	0.004	0.004	0	0	528
Study4	C	0.50	no		0.348	0.003	0.003	0	0	528
Study4	SM	0.05	no		0.309	0.035	0.043	33	63	528
Study4	SM	0.10	no		0.309	0.066	0.071	228	241	528
Study4	SM	0.20	no		0.309	0.108	0.110	266	158	528
Study4	SM	0.30	no		0.309	0.124	0.122	277	307	528
Study4	SM	0.50	no		0.309	0.121	0.115	282	273	528
V2	C	0.05	yes		0.364	0.015	0.015	383	382	109
V2	C	0.10	no		0.364	0.113	0.113	245	249	371
V2	C	0.20	no		0.364	0.425	0.426	369	368	528
V2	C	0.30	no		0.364	0.641	0.642	350	352	528
V2	C	0.50	no		0.364	0.578	0.578	331	329	528
V2	SM	0.05	yes		0.364	0.000	0.000	31	31	31
V2	SM	0.10	yes		0.364	0.010	0.010	123	123	123
V2	SM	0.20	yes		0.364	0.619	0.622	341	341	330
V2	SM	0.30	yes		0.364	0.990	0.990	443	448	425
V2	SM	0.50	no		0.364	0.879	0.877	359	358	367
V3	C	0.05	no		0.364	0.003	0.003	21	21	528
V3	C	0.10	no		0.364	0.007	0.007	0	0	528
V3	C	0.20	no		0.364	0.006	0.006	259	5	528
V3	C	0.30	no		0.364	0.006	0.006	0	0	528
V3	C	0.50	no		0.364	0.003	0.003	0	0	528
V3	SM	0.05	no		0.362	0.030	0.031	95	95	528
V3	SM	0.10	no		0.362	0.062	0.062	225	233	528
V3	SM	0.20	no		0.362	0.114	0.113	288	336	528
V3	SM	0.30	no		0.362	0.134	0.134	255	256	528
V3	SM	0.50	no		0.362	0.148	0.148	337	339	528
V4	C	0.05	no		0.362	0.058	0.058	211	212	314
V4	C	0.10	no		0.362	0.152	0.152	293	278	443
V4	C	0.20	no		0.362	0.370	0.373	347	326	528
V4	C	0.30	no		0.362	0.478	0.481	459	445	528
V4	C	0.50	no		0.362	0.429	0.423	383	376	528
V4	SM	0.05	no		0.344	0.000	0.000	0	0	528
V4	SM	0.10	no		0.344	0.000	0.000	0	0	528
V4	SM	0.20	no		0.344	0.000	0.000	0	0	528
V4	SM	0.30	no		0.344	0.000	0.000	0	0	528
V4	SM	0.50	no		0.344	0.000	0.000	0	0	528